

BFS Traversal

Step 1: Start at A

Legend

Complete

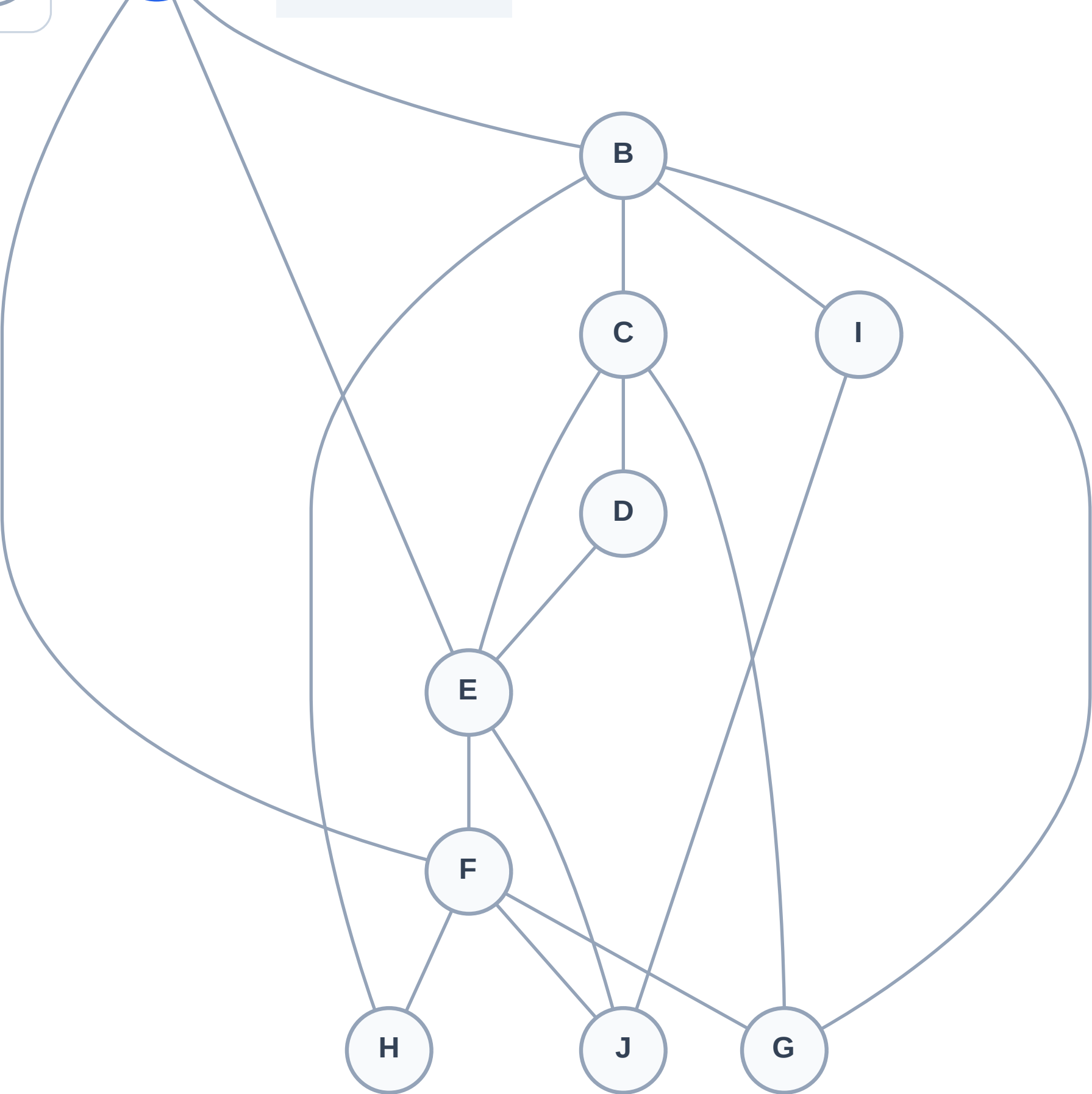
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

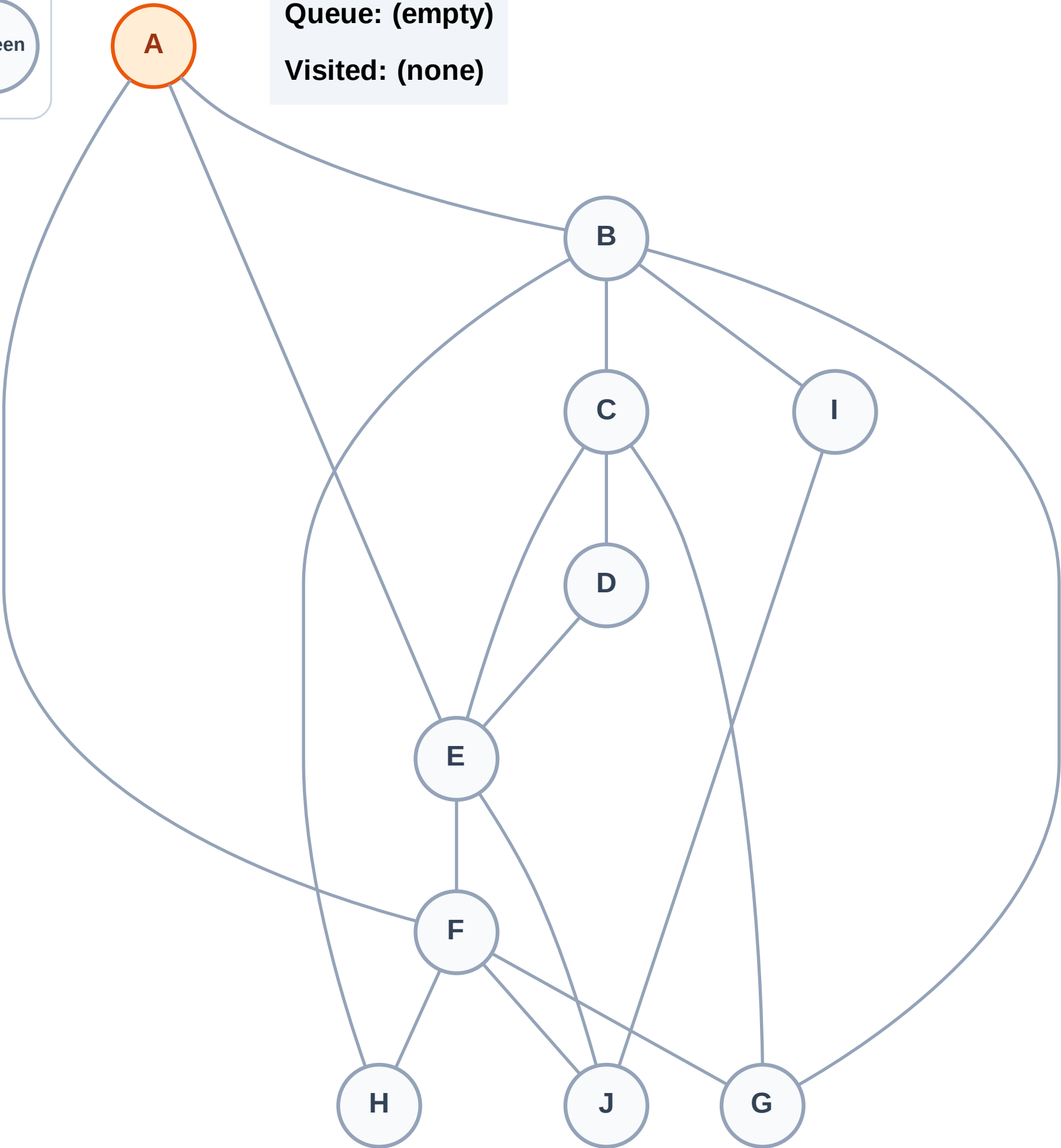
Complete

Processing

Discovered

Unseen

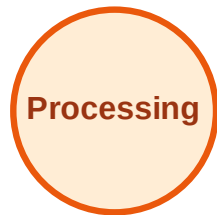
Queue: (empty)
Visited: (none)



BFS Traversal

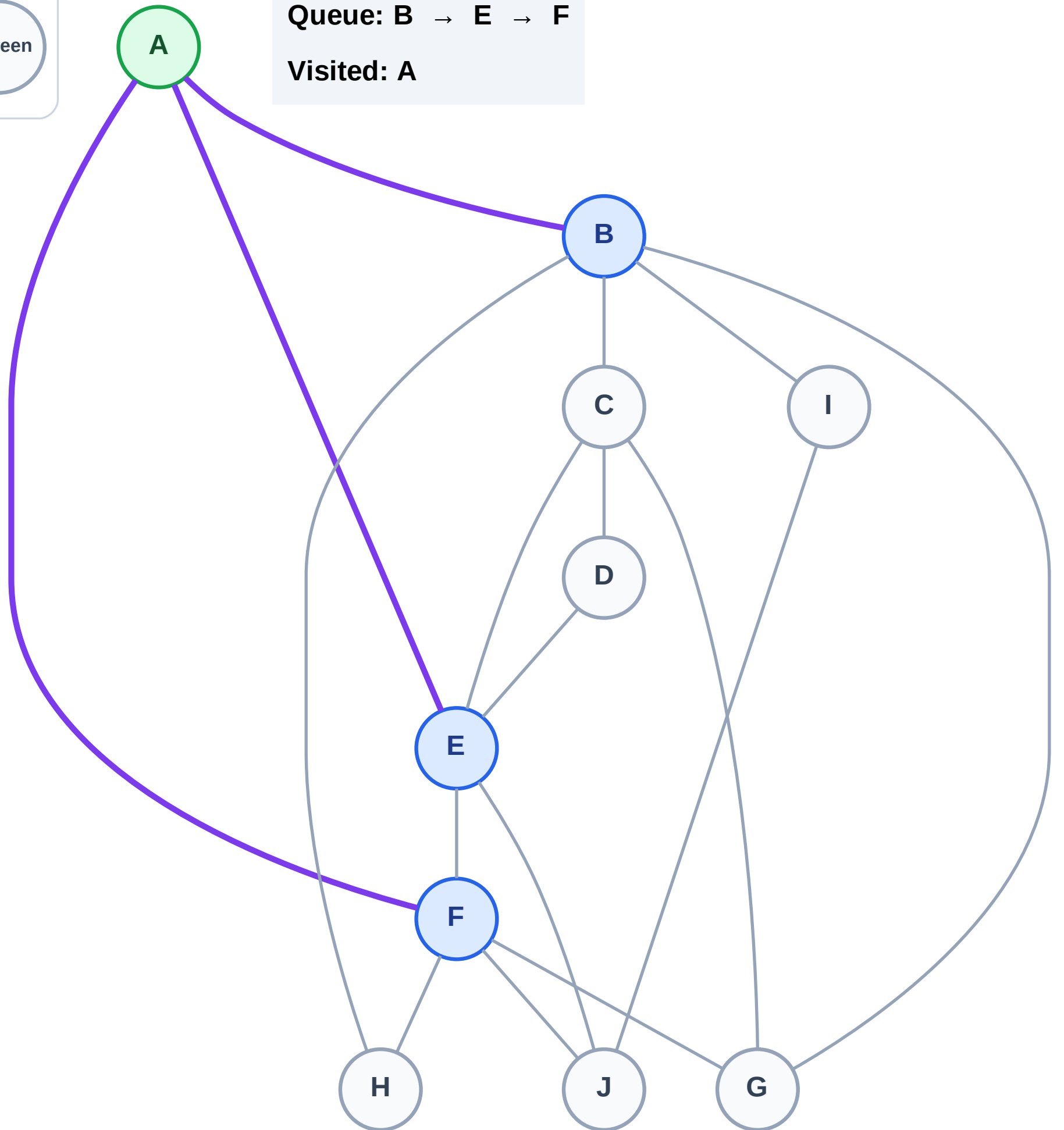
Step 3: Discover B, E, F from A

Legend



Queue: B → E → F

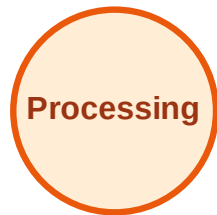
Visited: A



BFS Traversal

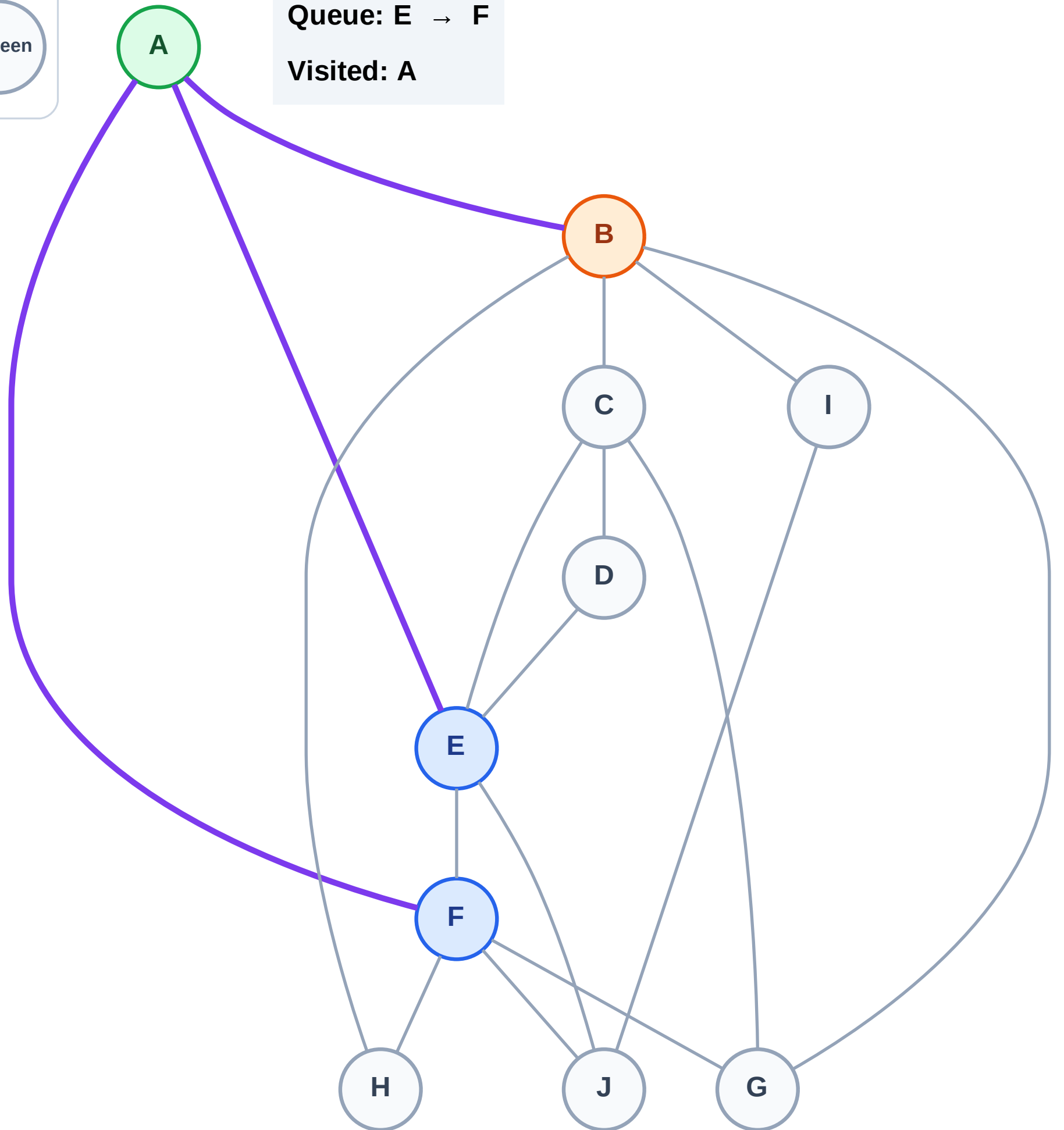
Step 4: Process node B

Legend



Queue: E → F

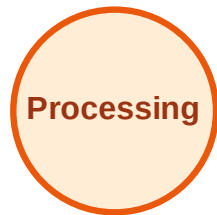
Visited: A



BFS Traversal

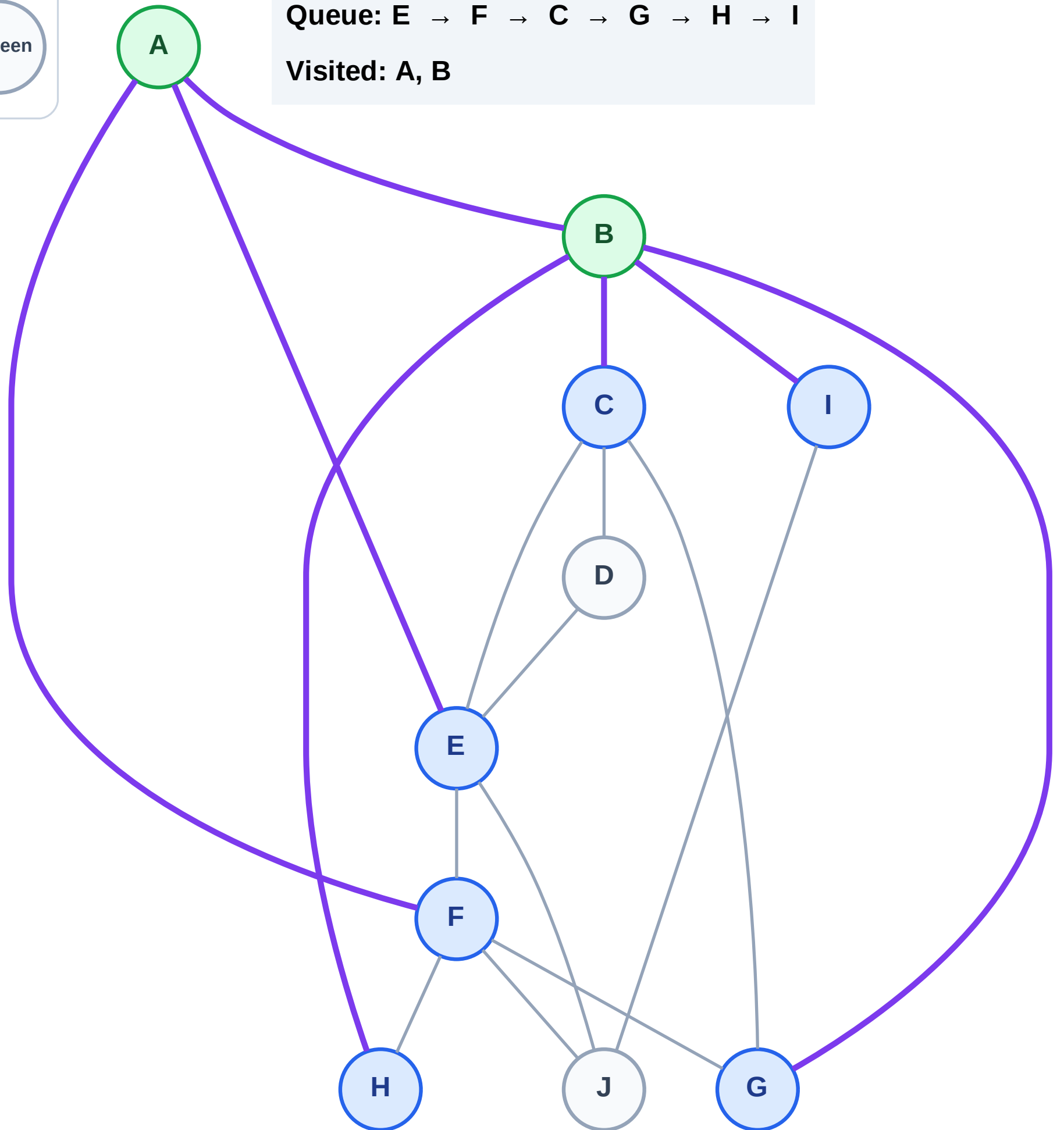
Step 5: Discover C, G, H, I from B

Legend



Queue: E → F → C → G → H → I

Visited: A, B



BFS Traversal

Step 6: Process node E

Legend

Complete

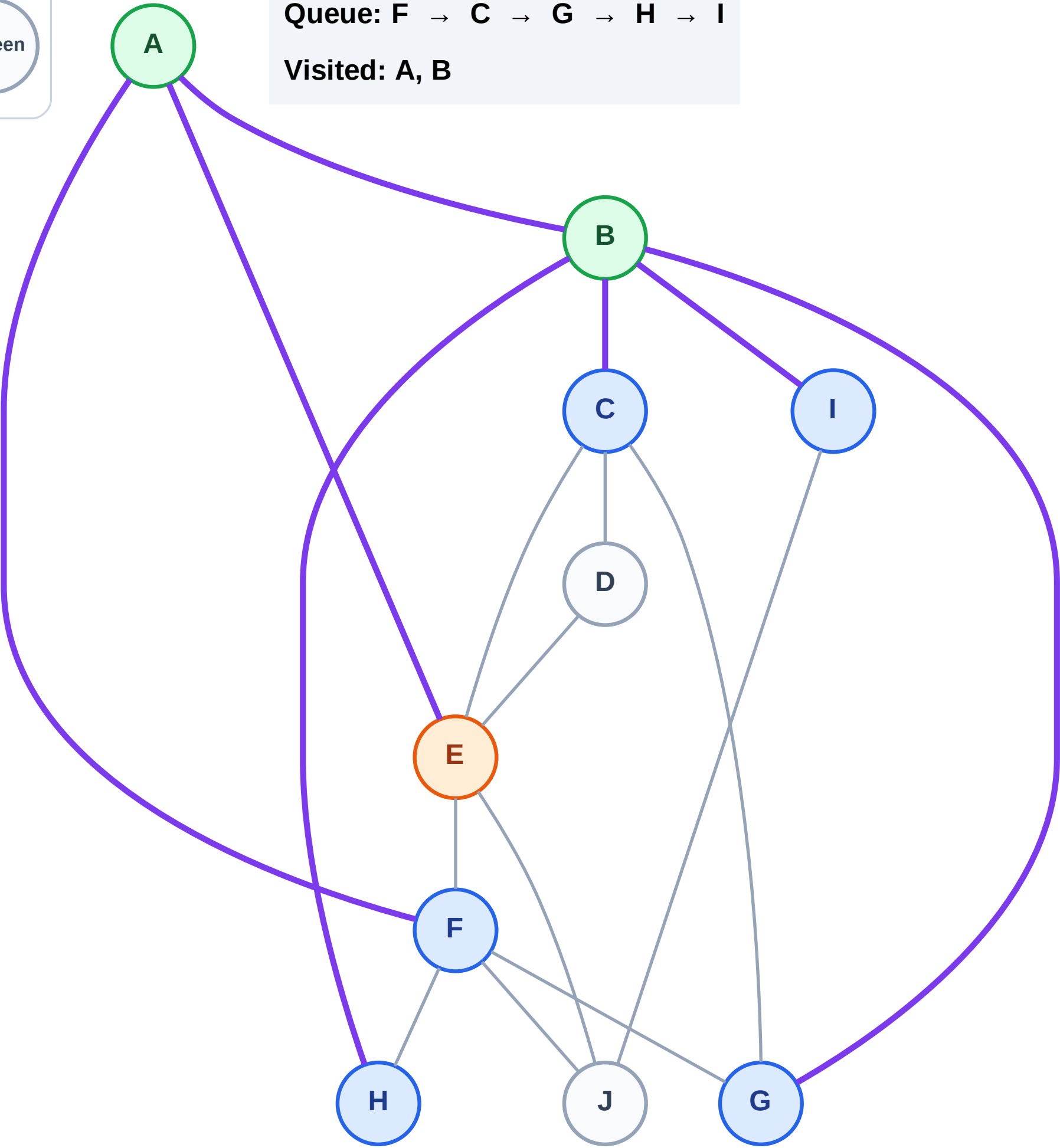
Processing

Discovered

Unseen

Queue: F → C → G → H → I

Visited: A, B



Step 7: Discover D, J from E

Queue: F → C → G → H → I → D → J

Complete

Processing

Discovered

Unseen



F

Н

J

G

BFS Traversal

Step 8: Process node F

Legend

Complete

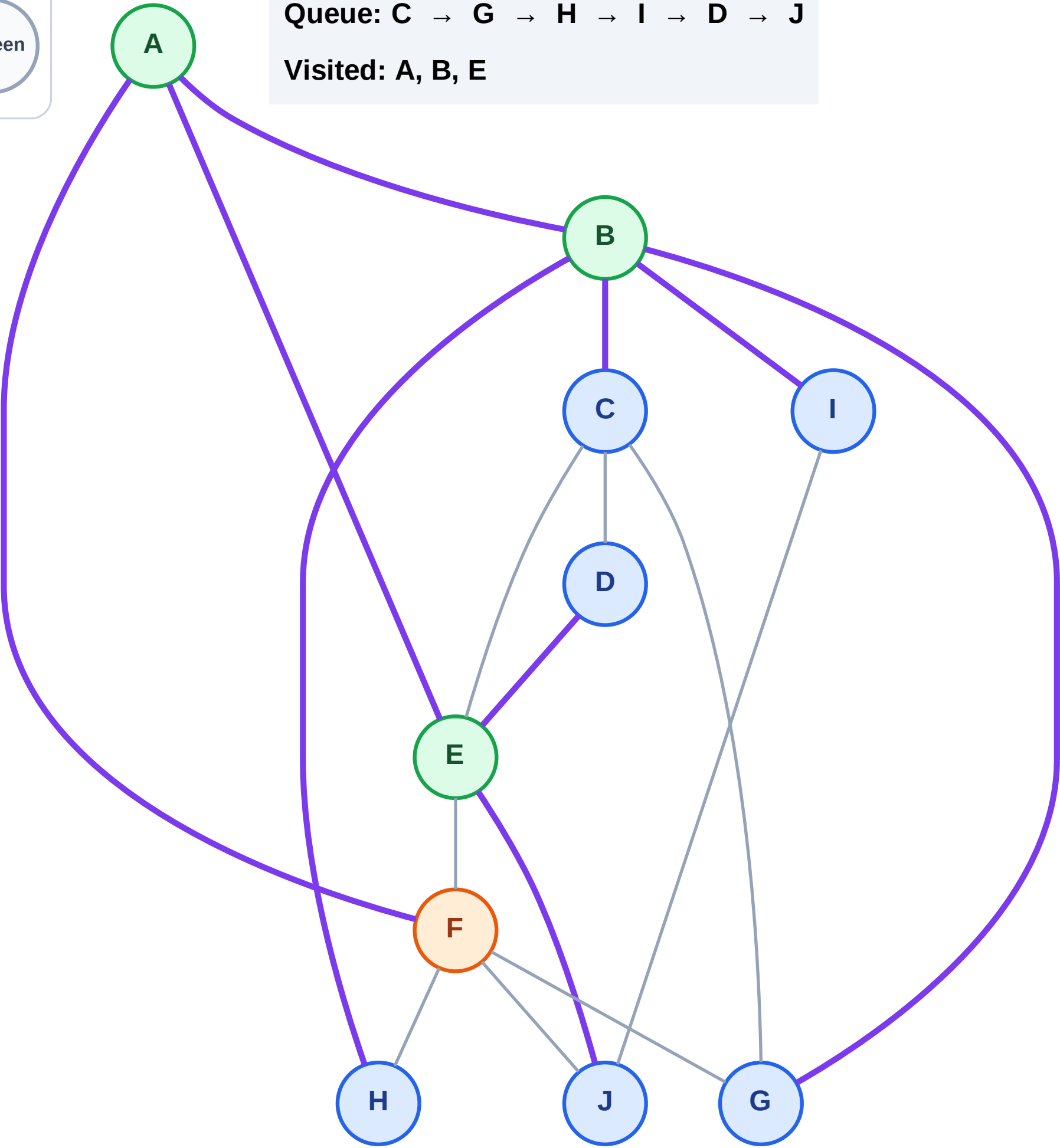
Processing

Discovered

Unseen

Queue: C → G → H → I → D → J

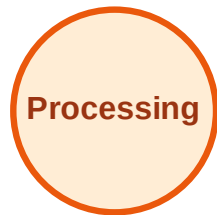
Visited: A, B, E



BFS Traversal

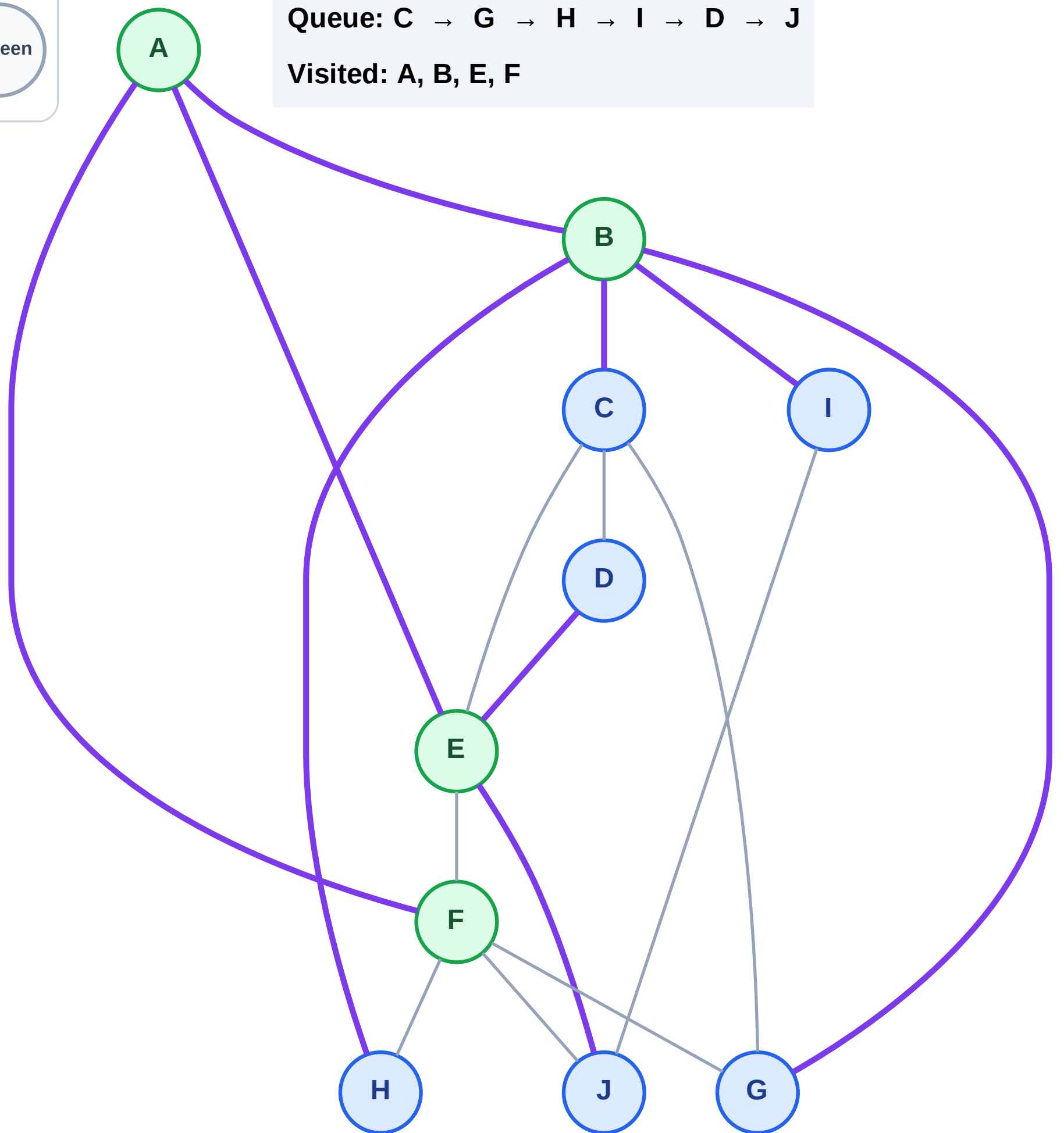
Step 9: No new neighbors from F

Legend



Queue: C → G → H → I → D → J

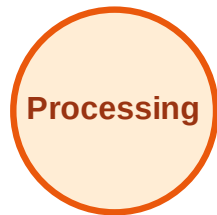
Visited: A, B, E, F



BFS Traversal

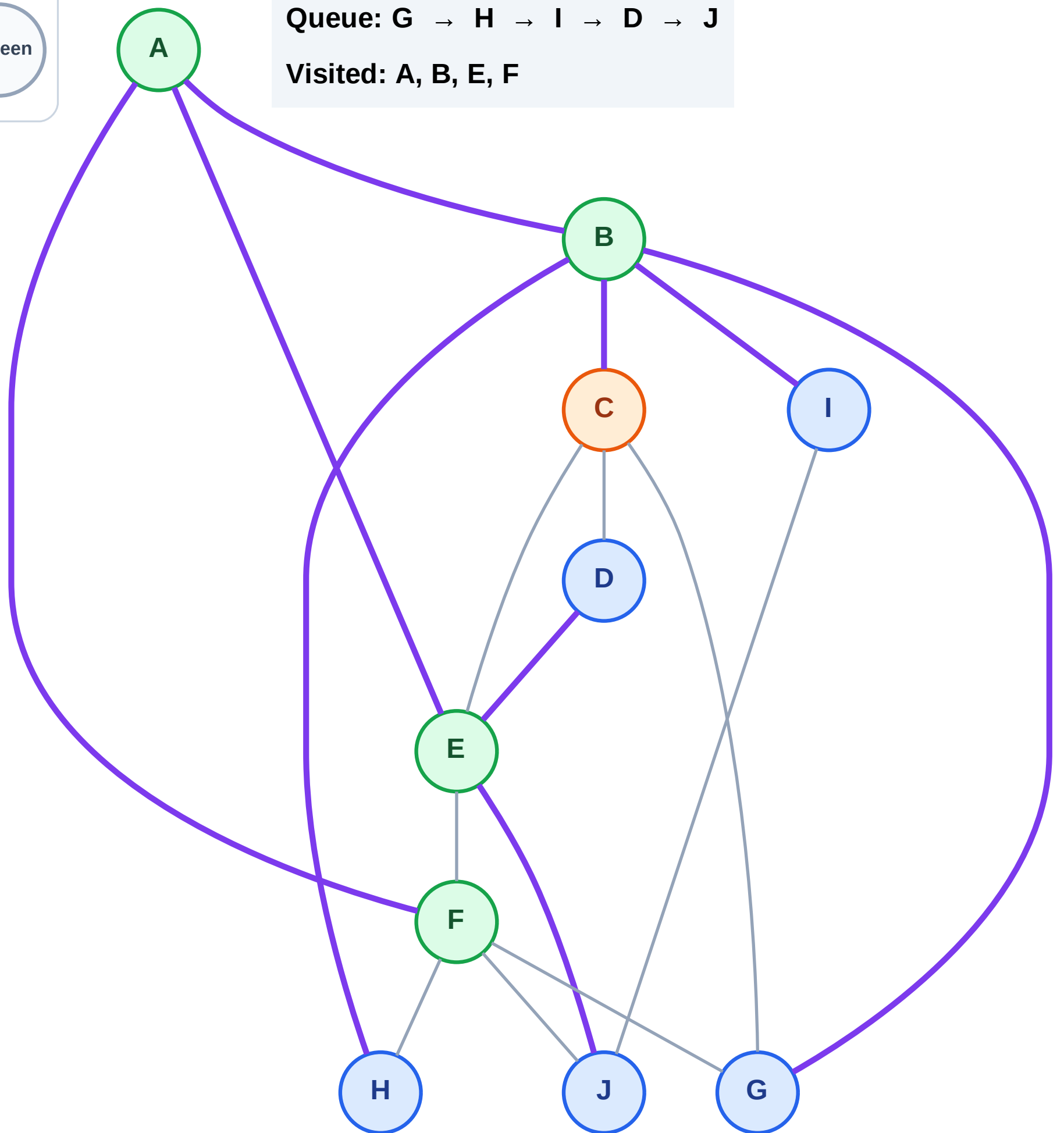
Step 10: Process node C

Legend



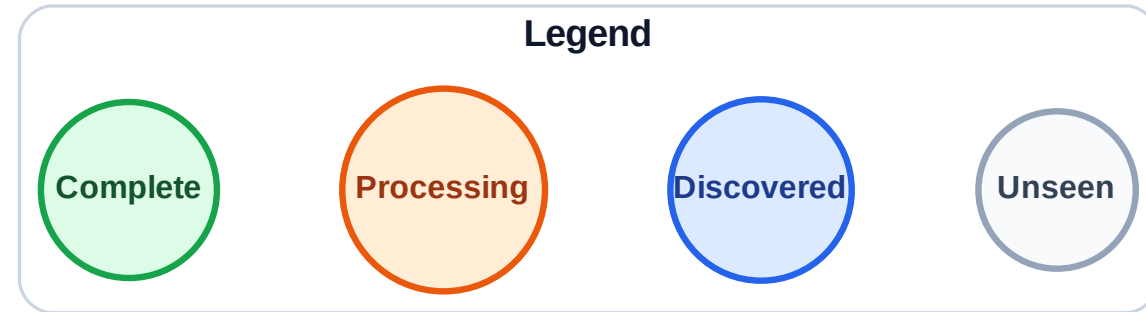
Queue: G → H → I → D → J

Visited: A, B, E, F



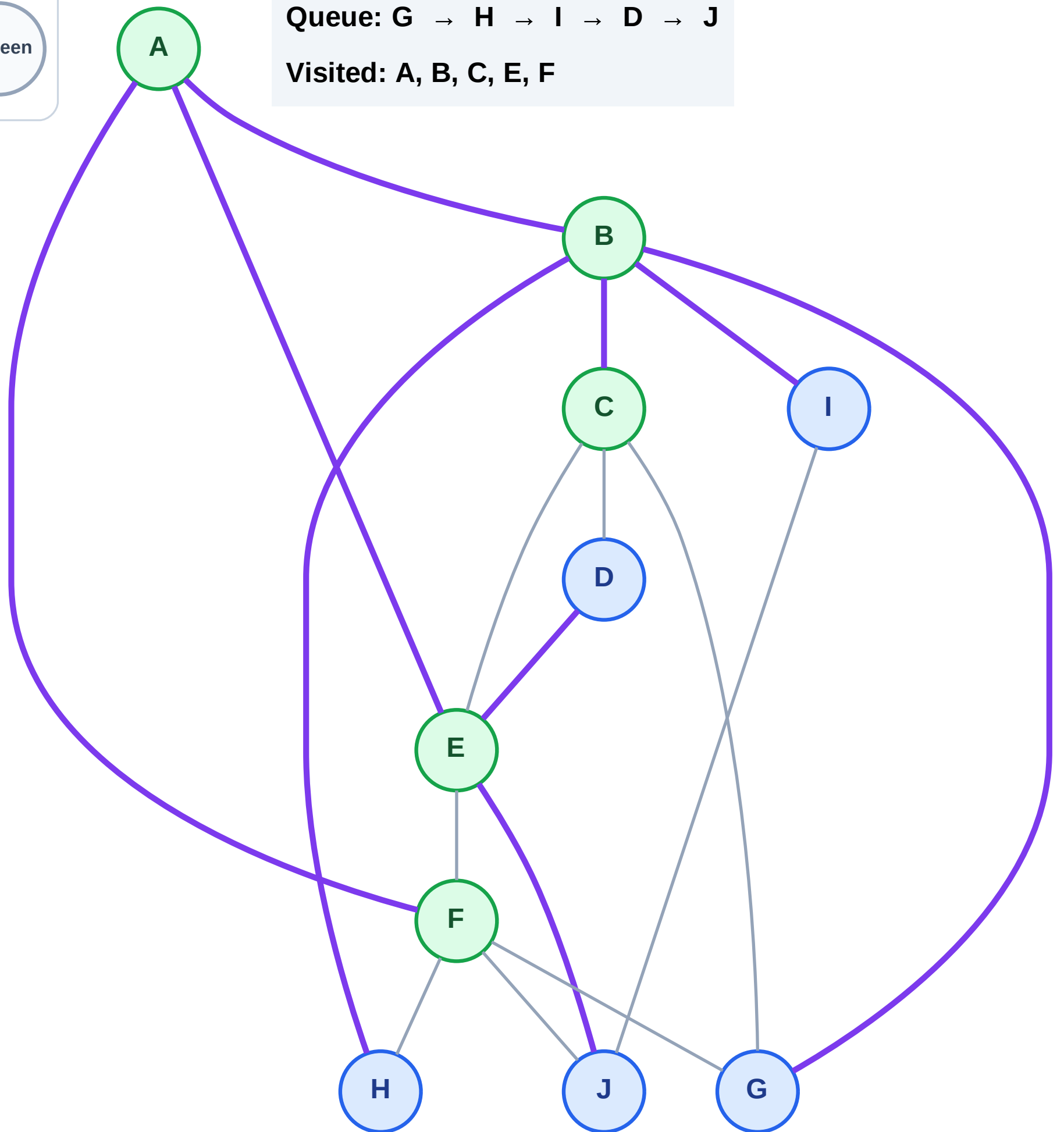
BFS Traversal

Step 11: No new neighbors from C



Queue: G → H → I → D → J

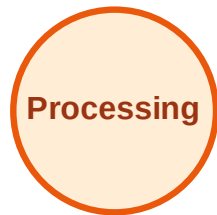
Visited: A, B, C, E, F



BFS Traversal

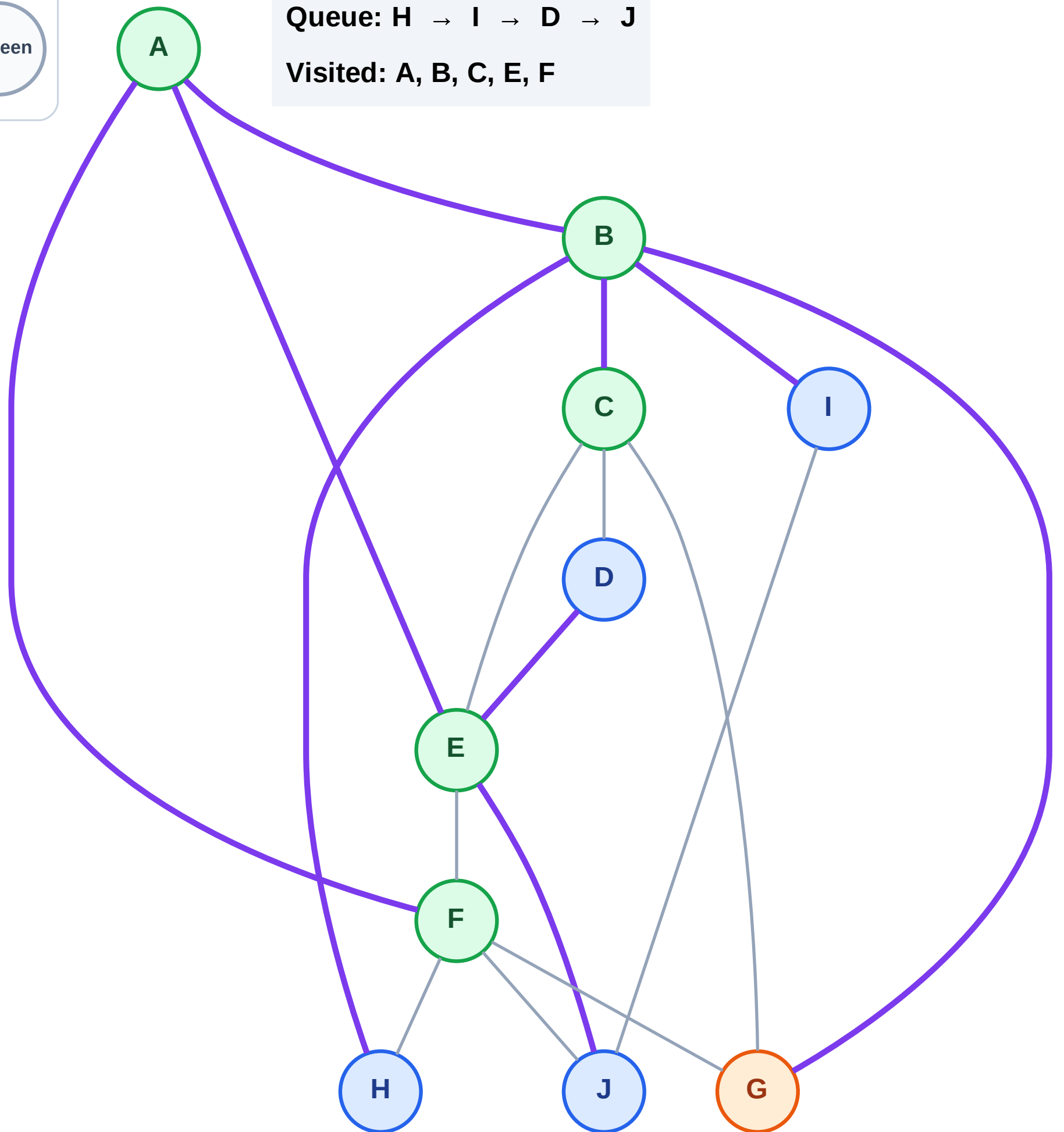
Step 12: Process node G

Legend



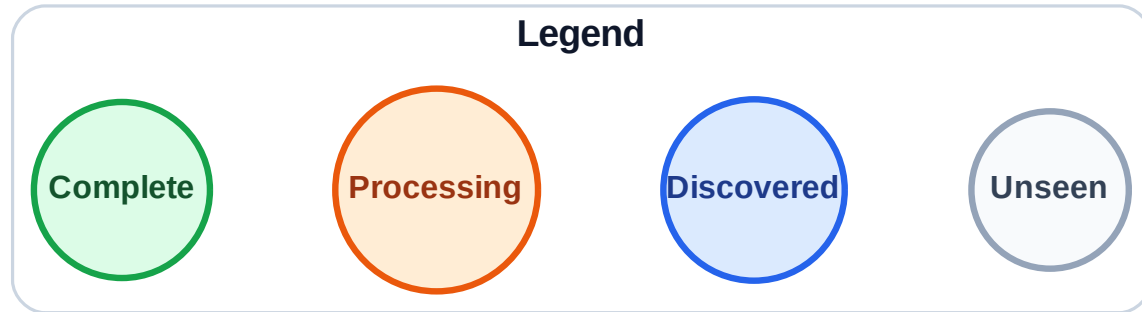
Queue: H → I → D → J

Visited: A, B, C, E, F



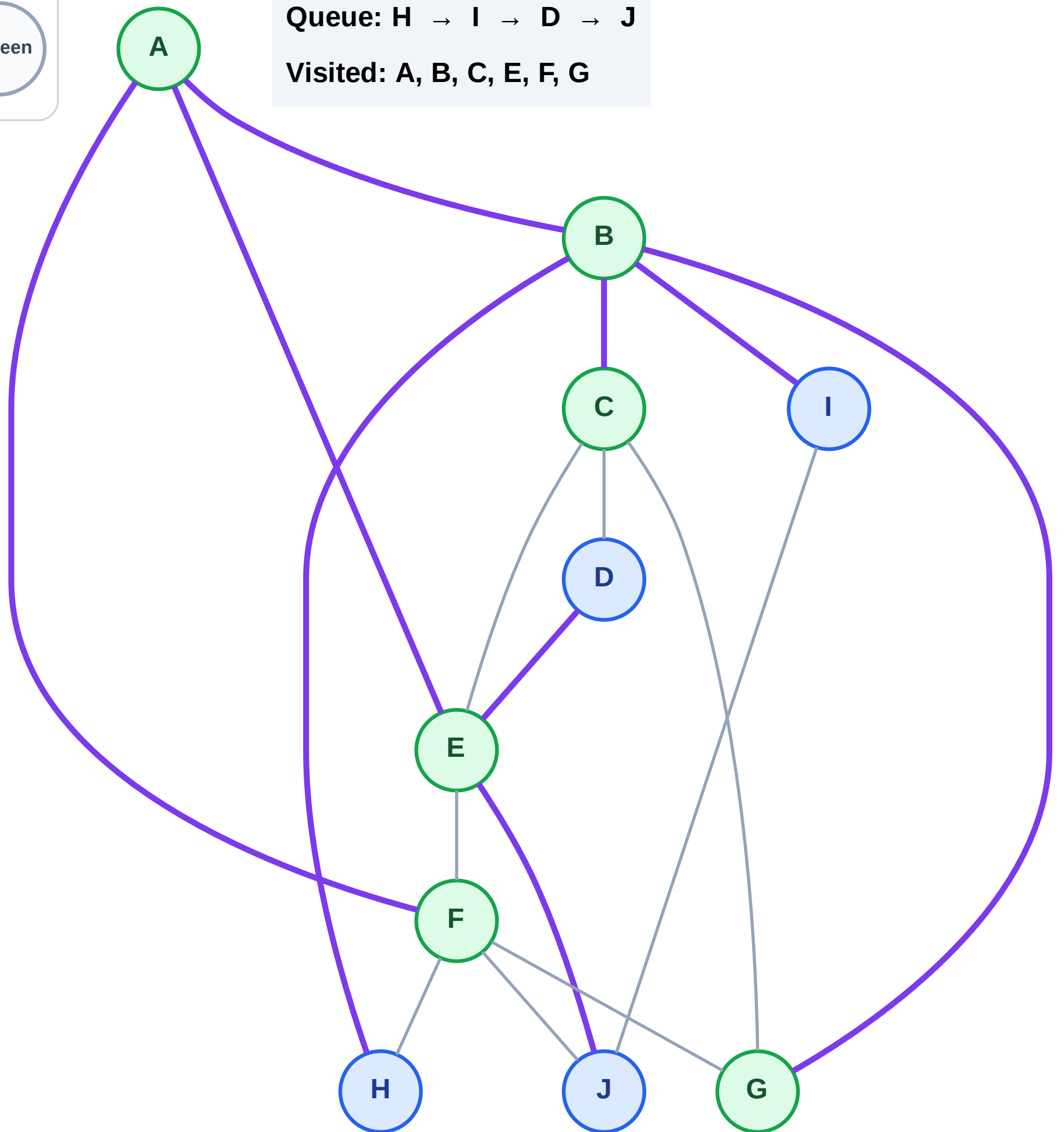
BFS Traversal

Step 13: No new neighbors from G



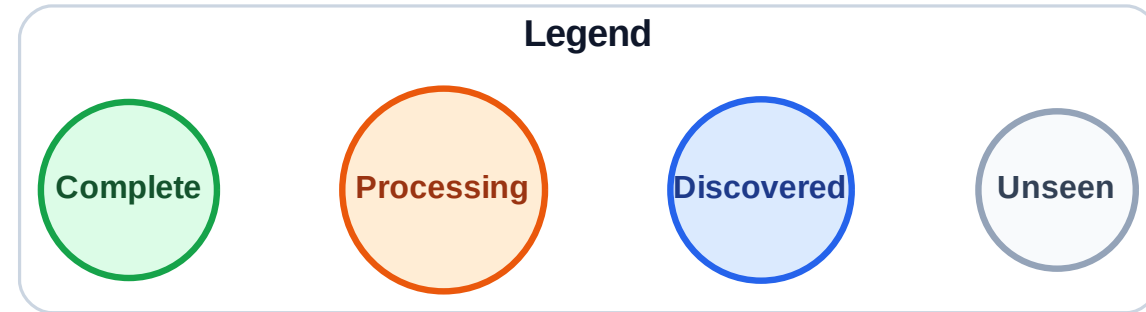
Queue: H → I → D → J

Visited: A, B, C, E, F, G



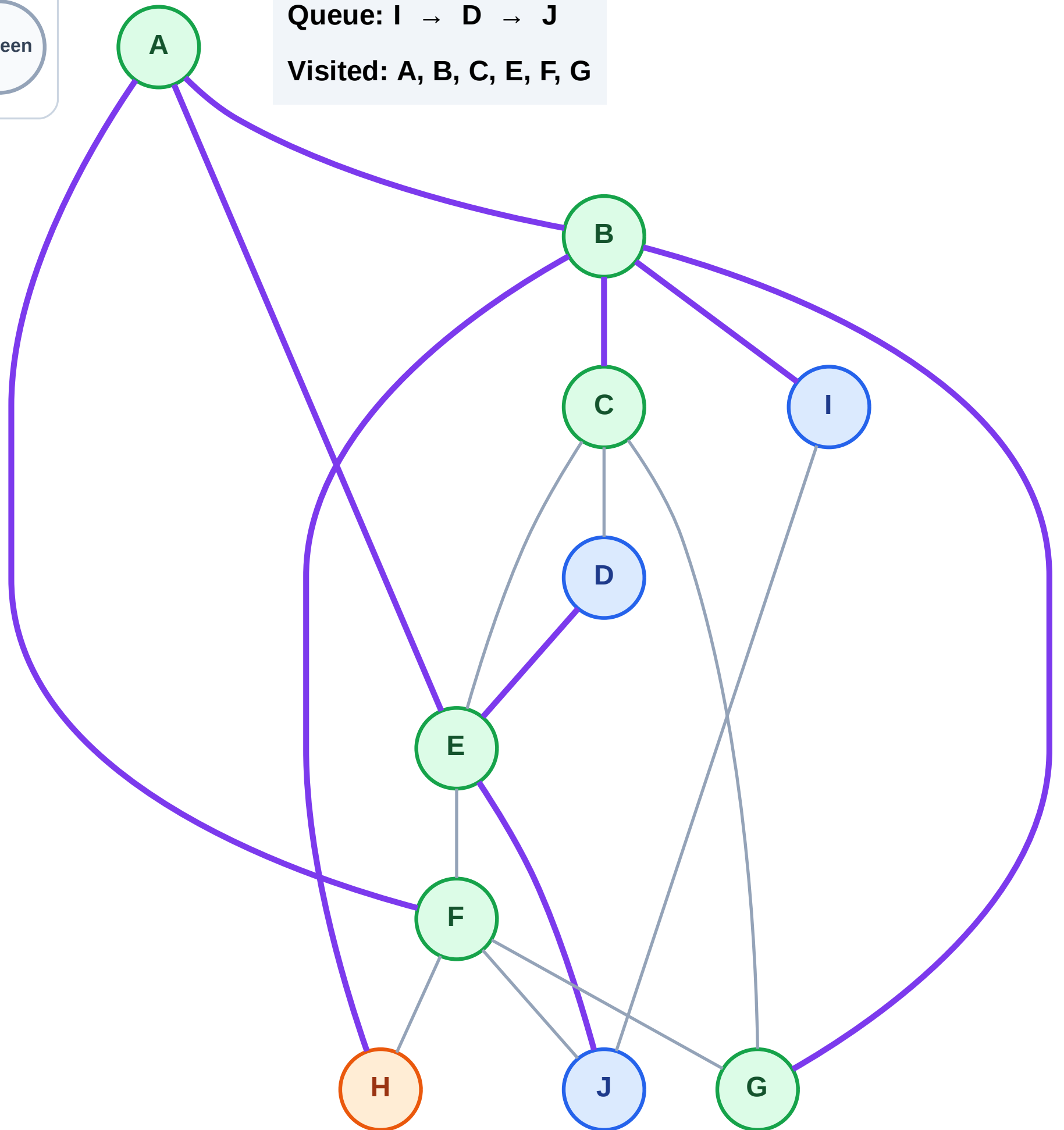
BFS Traversal

Step 14: Process node H



Queue: I → D → J

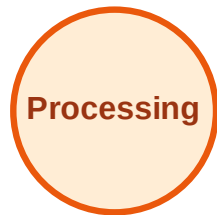
Visited: A, B, C, E, F, G



BFS Traversal

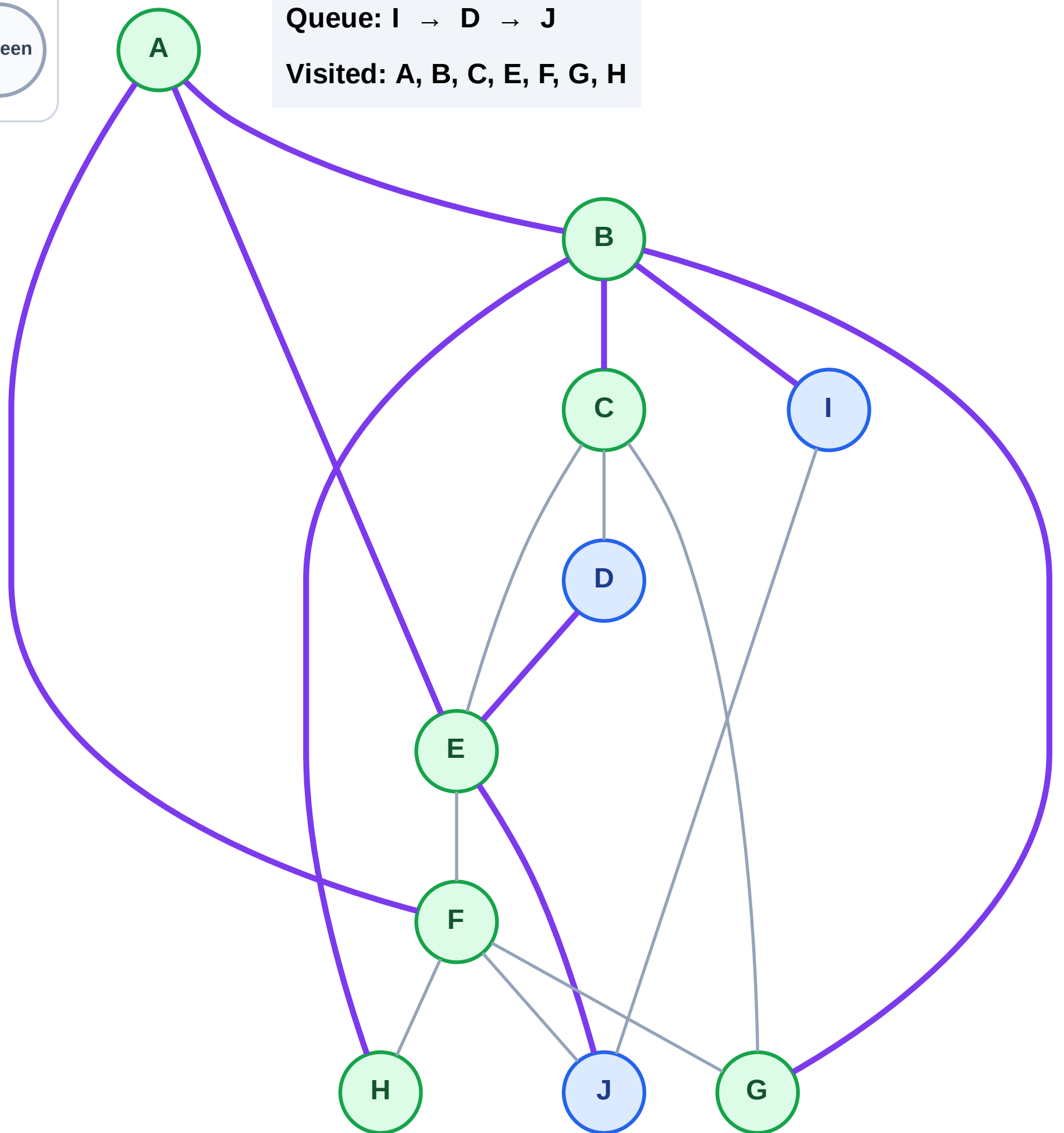
Step 15: No new neighbors from H

Legend



Queue: I → D → J

Visited: A, B, C, E, F, G, H



BFS Traversal

Step 16: Process node I

Legend

Complete

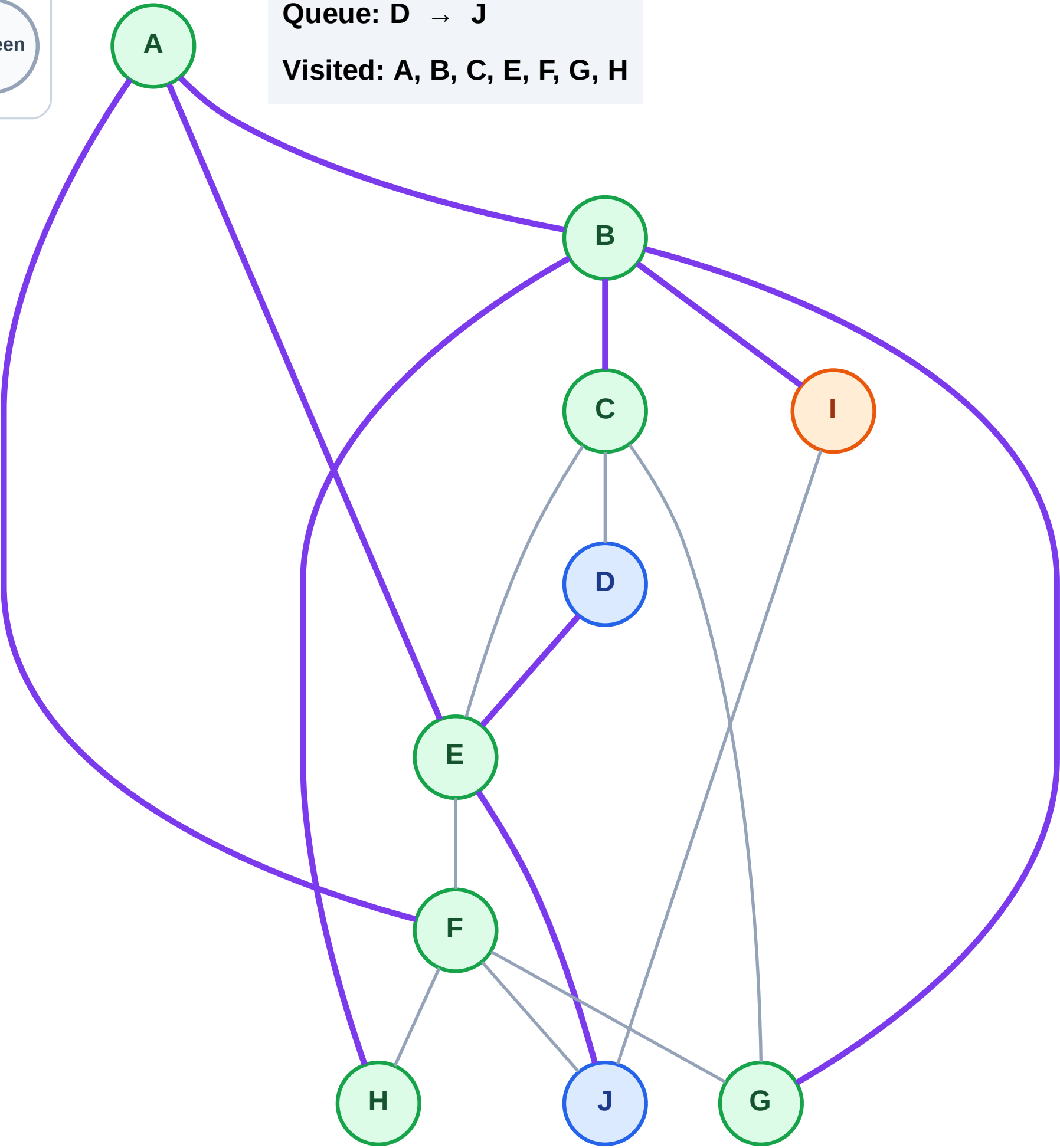
Processing

Discovered

Unseen

Queue: D → J

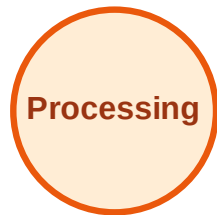
Visited: A, B, C, E, F, G, H



BFS Traversal

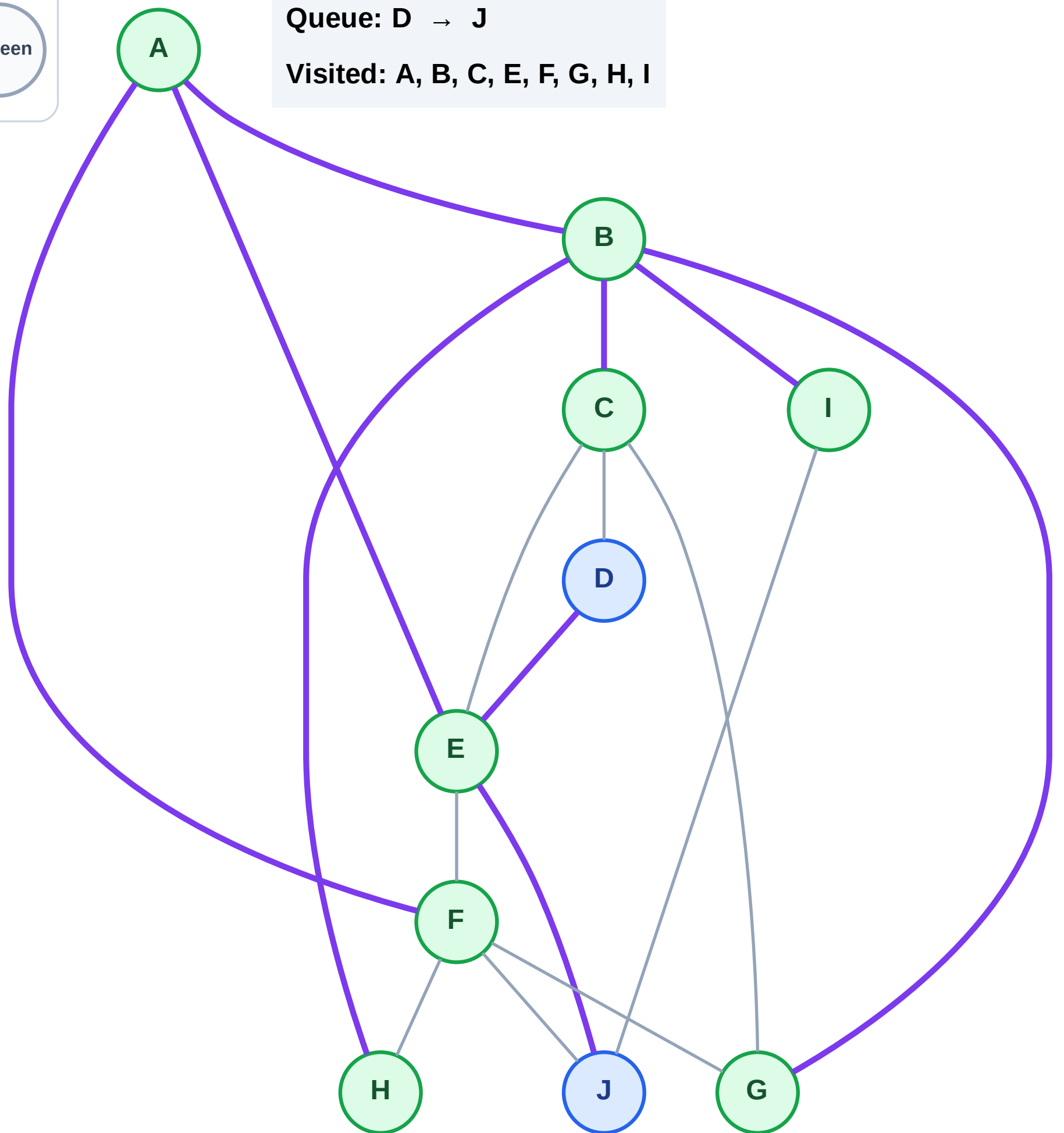
Step 17: No new neighbors from I

Legend



Queue: D → J

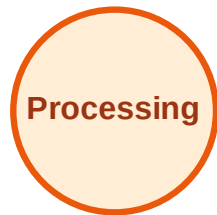
Visited: A, B, C, E, F, G, H, I



BFS Traversal

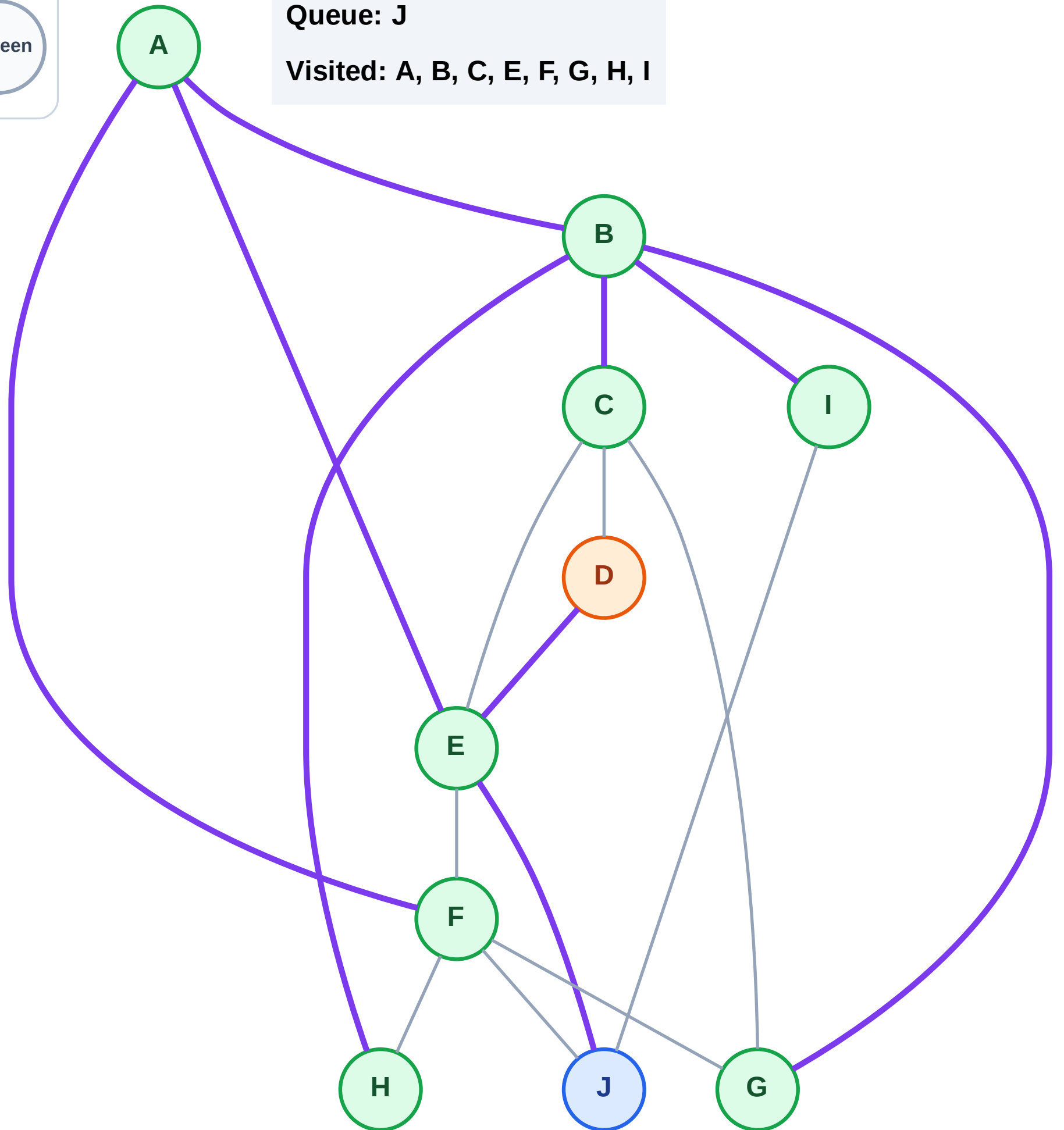
Step 18: Process node D

Legend



Queue: J

Visited: A, B, C, E, F, G, H, I



BFS Traversal

Step 19: No new neighbors from D

Legend

Complete

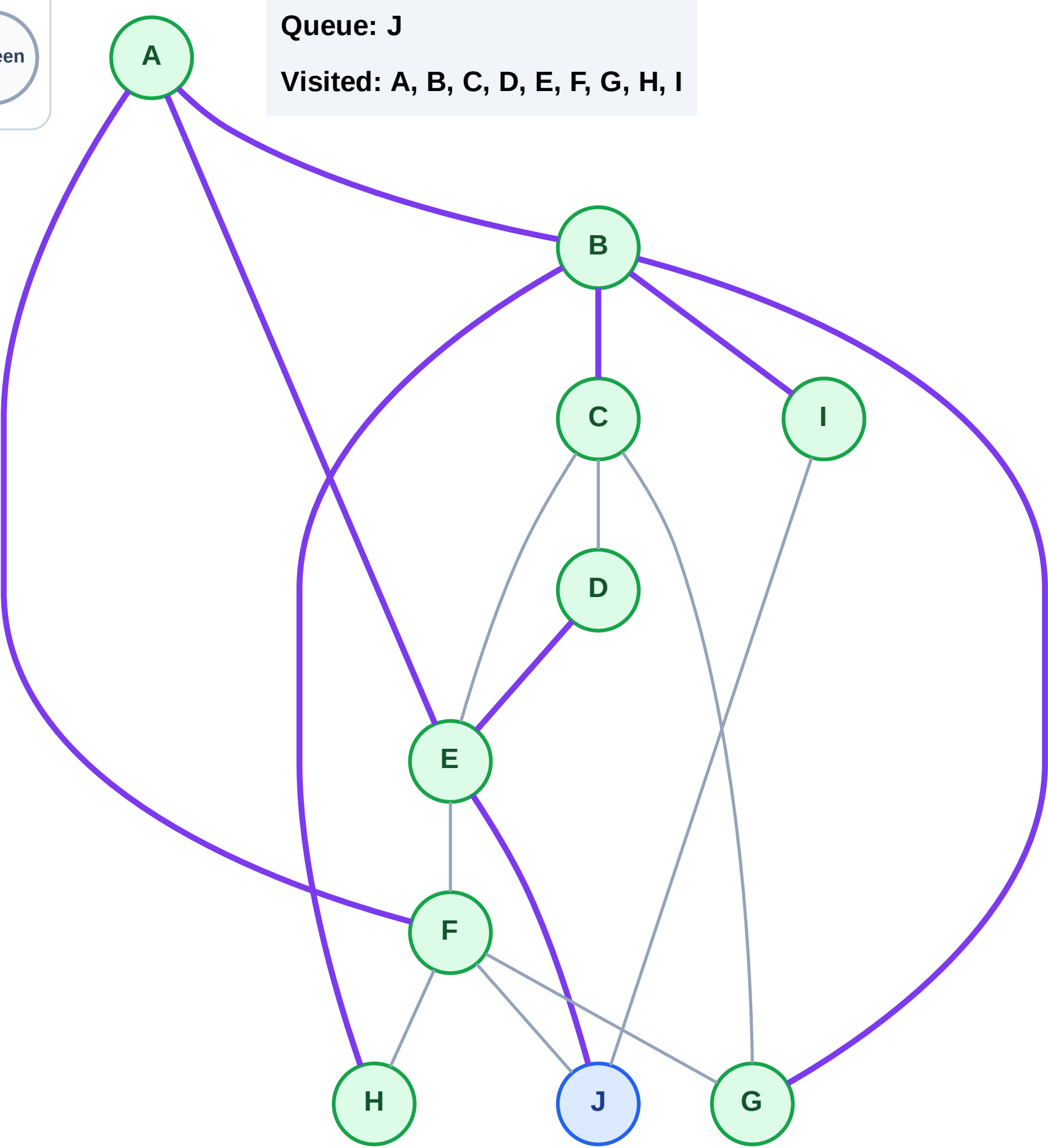
Processing

Discovered

Unseen

Queue: J

Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

Step 20: Process node J

Legend

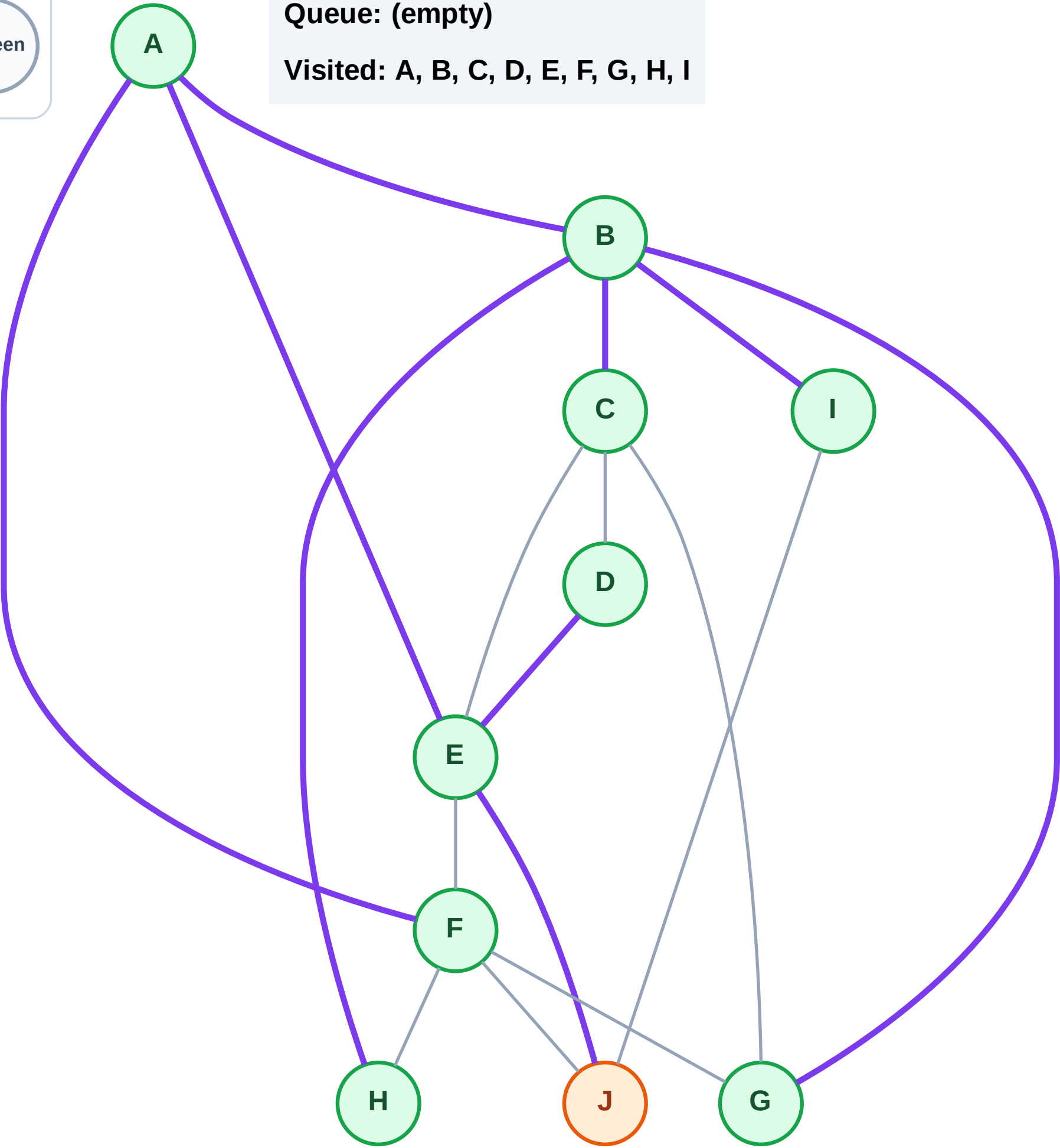
Complete

Processing

Discovered

Unseen

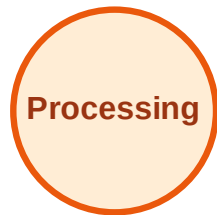
Queue: (empty)
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

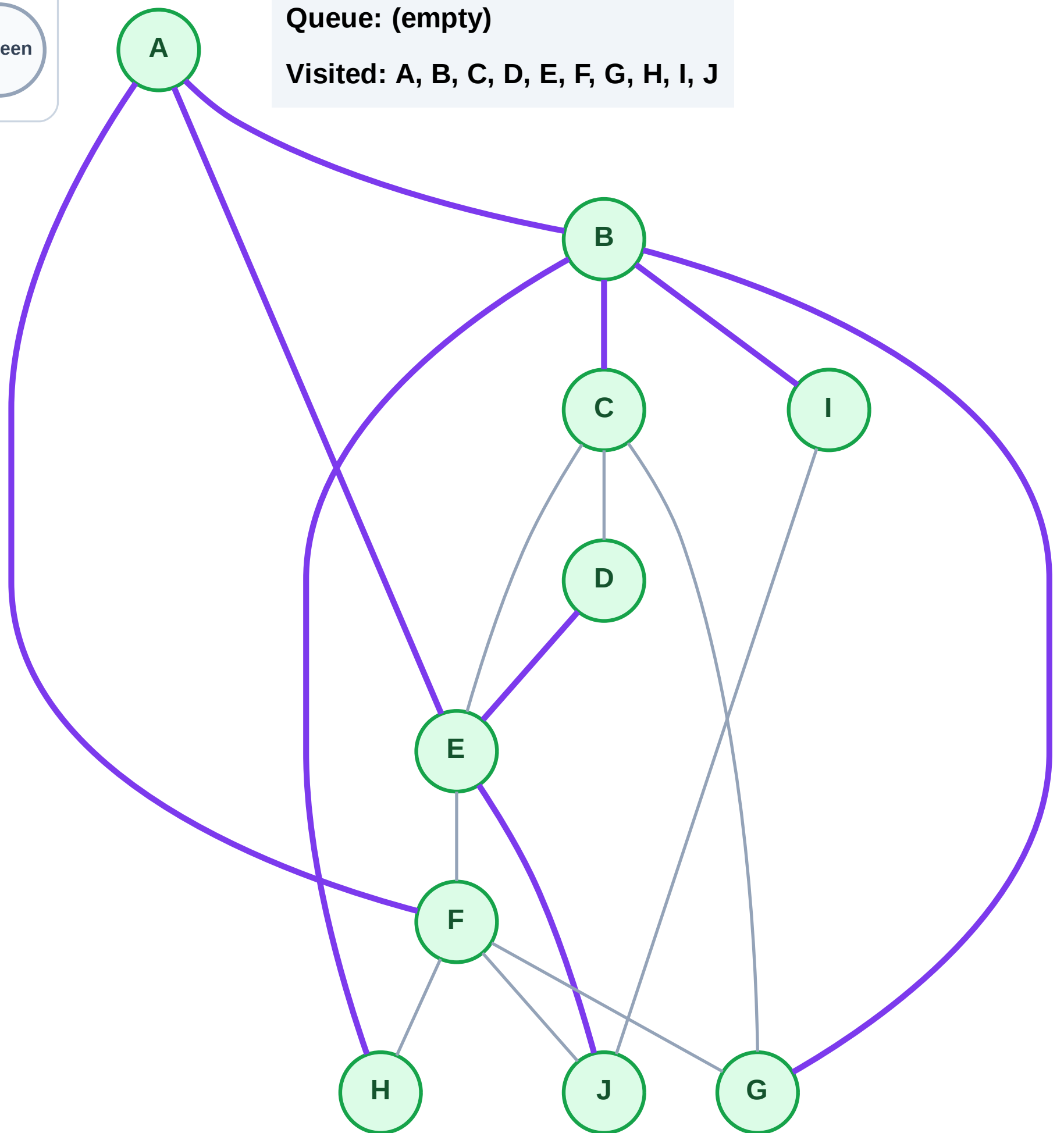
Step 21: No new neighbors from J

Legend



Queue: (empty)

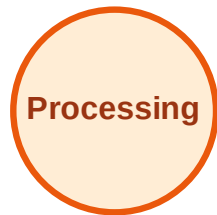
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

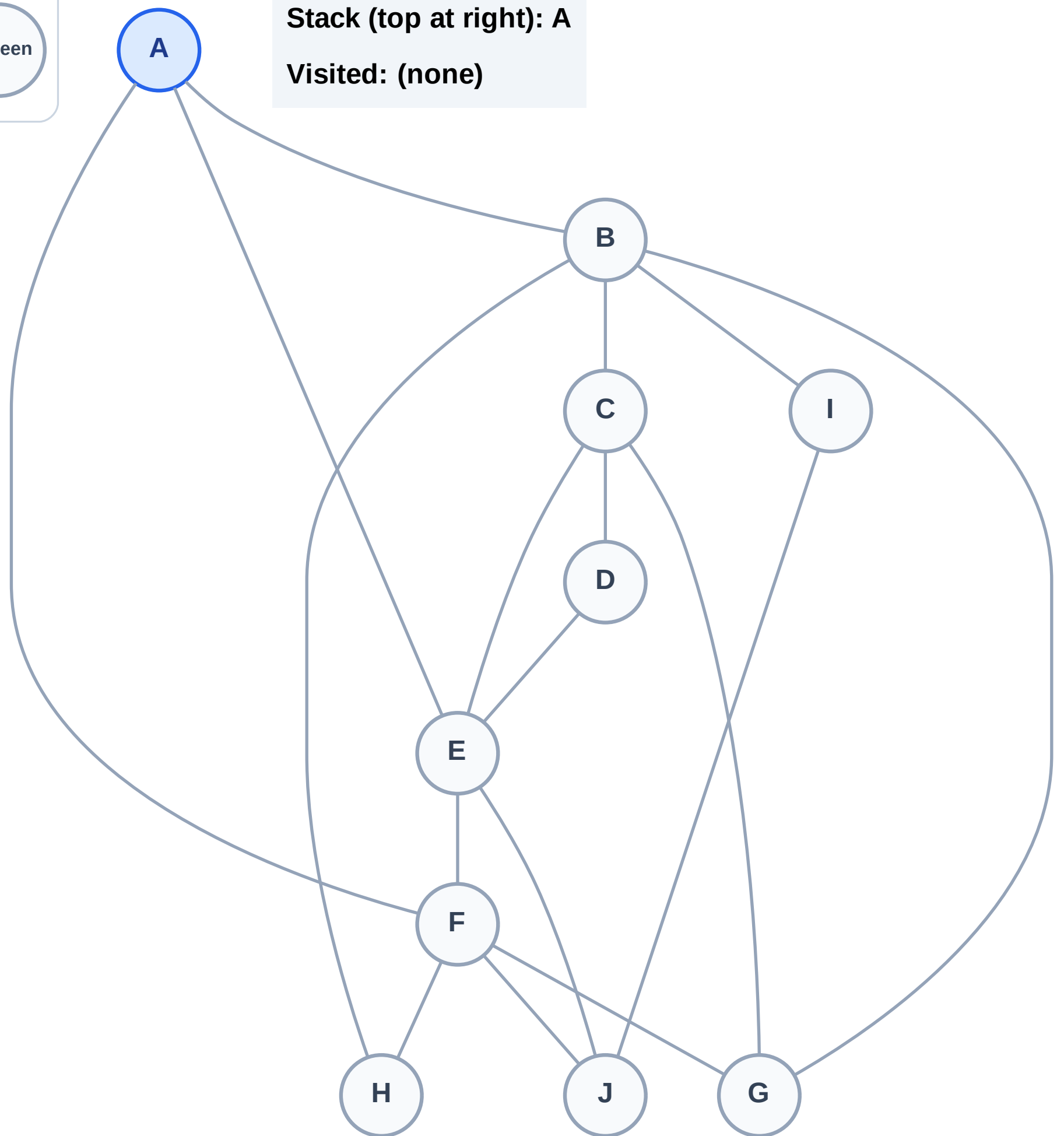
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

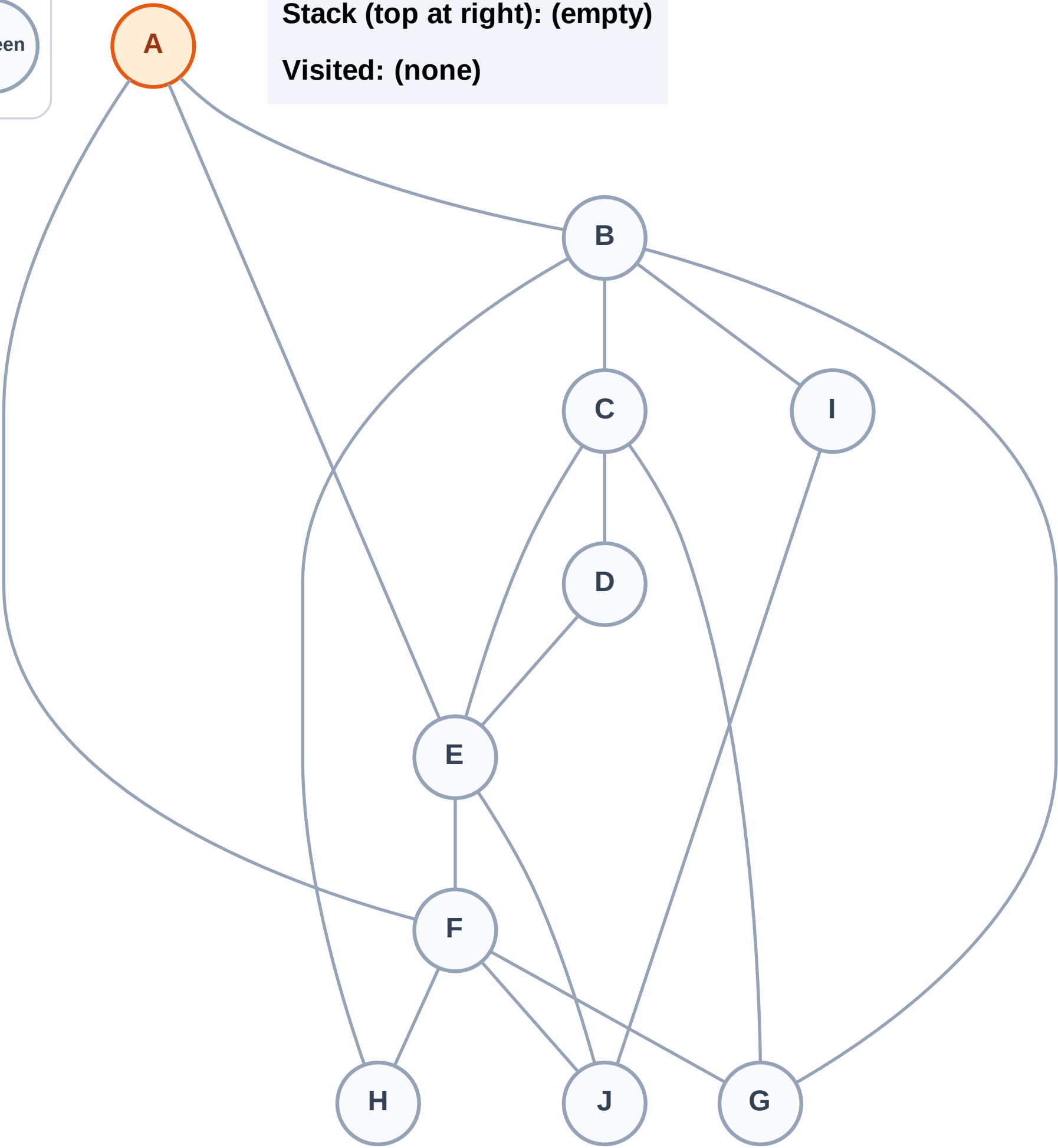
Complete

Processing

Discovered

Unseen

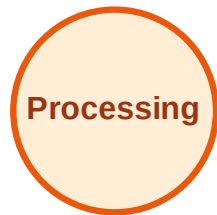
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

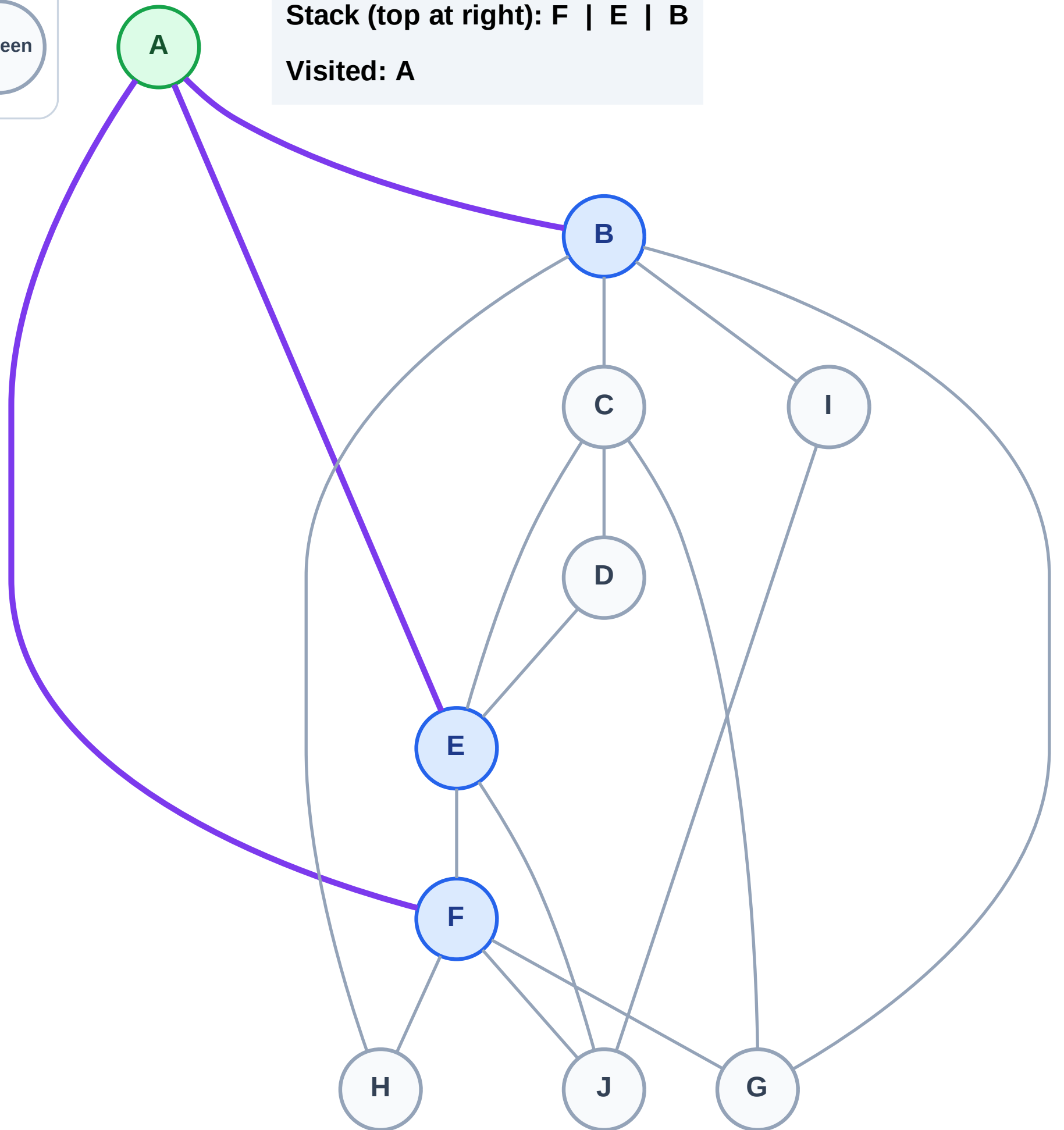
Step 3: Discover F, E, B from A

Legend



Stack (top at right): F | E | B

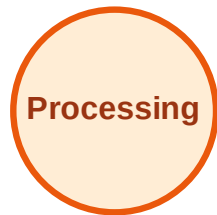
Visited: A



DFS Traversal

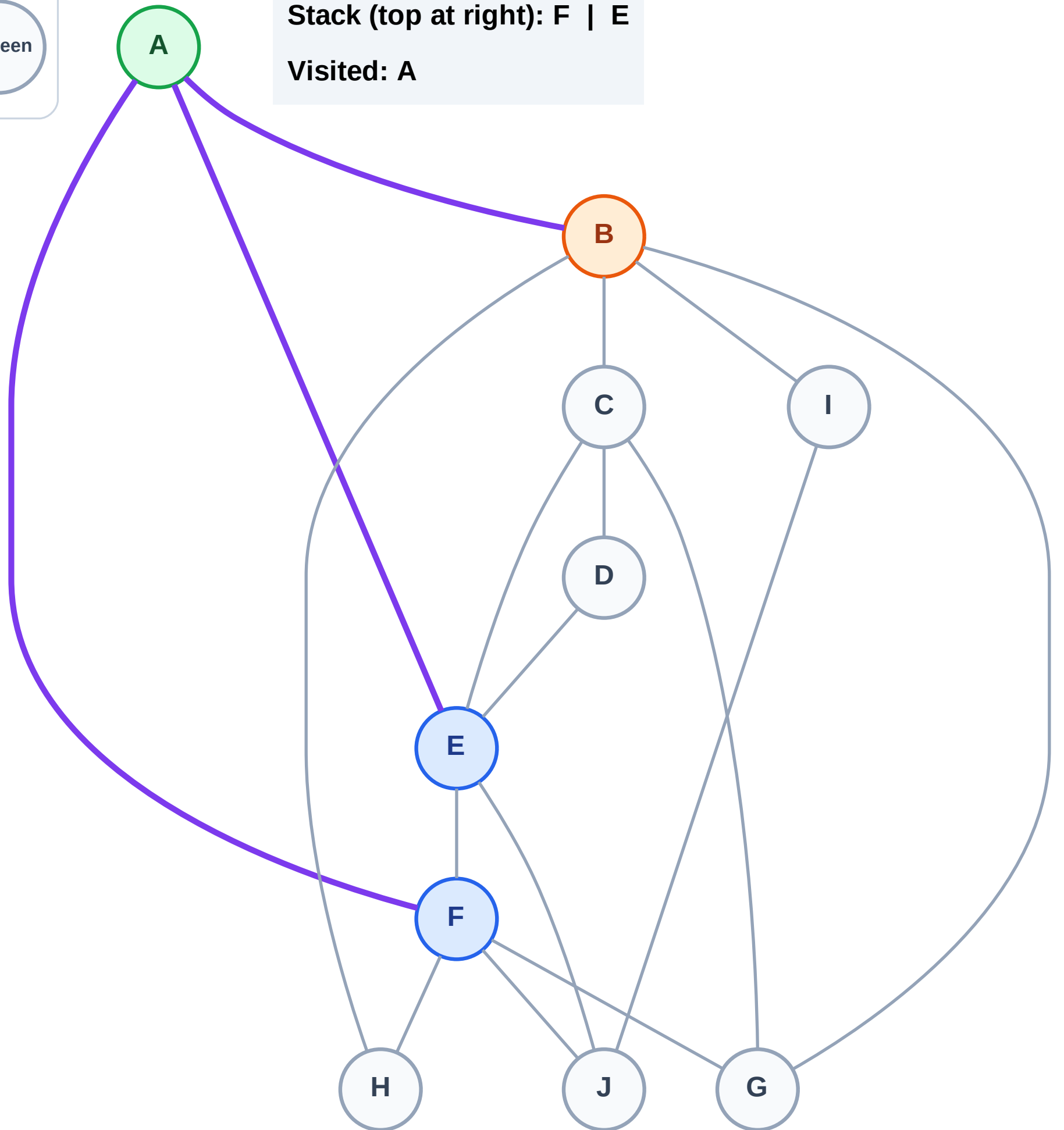
Step 4: Process node B

Legend



Stack (top at right): F | E

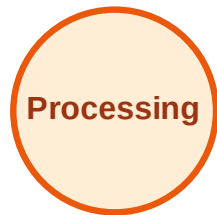
Visited: A



DFS Traversal

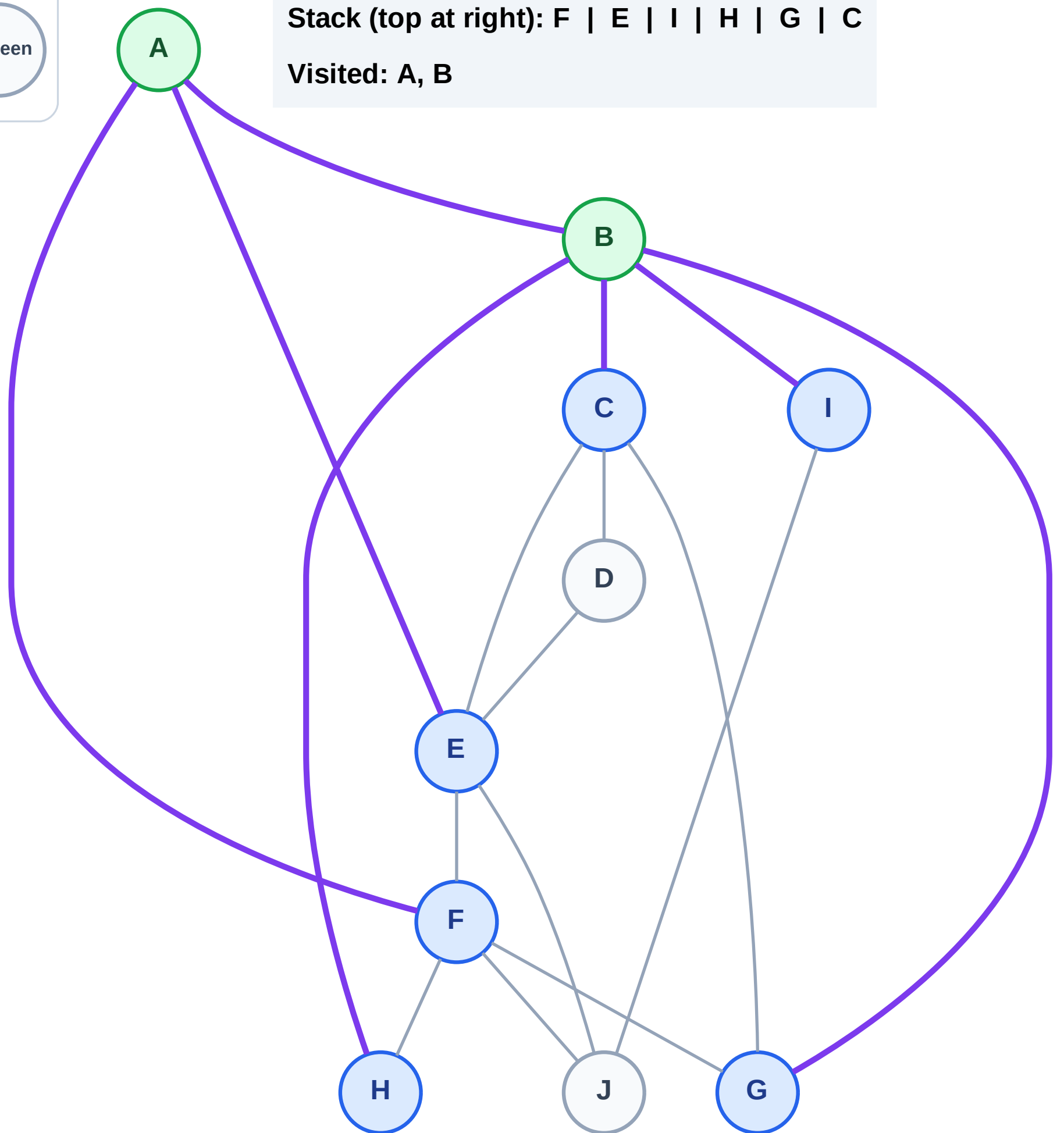
Step 5: Discover I, H, G, C from B

Legend



Stack (top at right): F | E | I | H | G | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

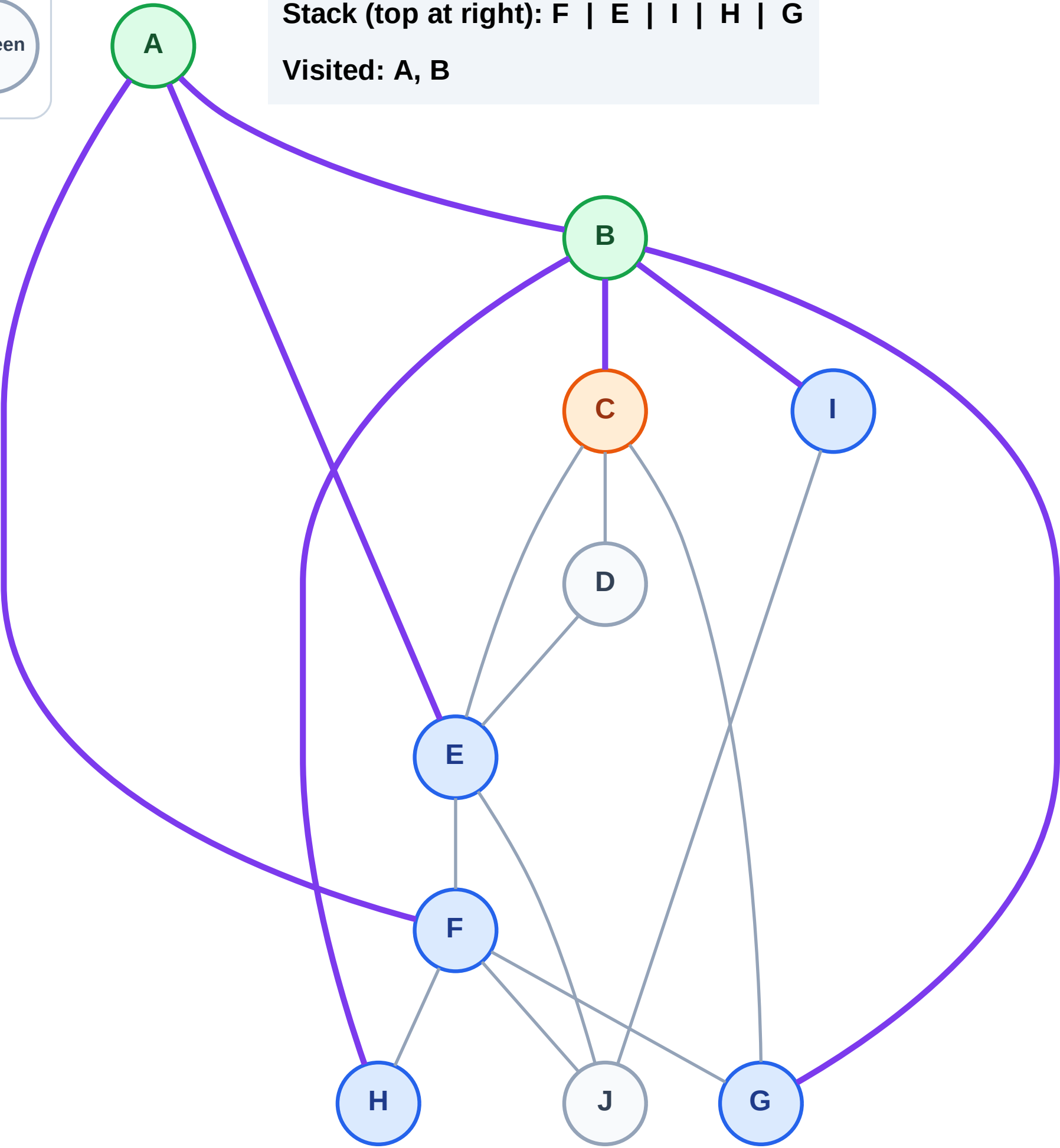
Processing

Discovered

Unseen

Stack (top at right): F | E | I | H | G

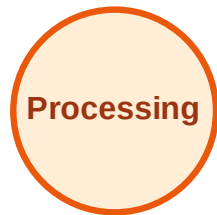
Visited: A, B



DFS Traversal

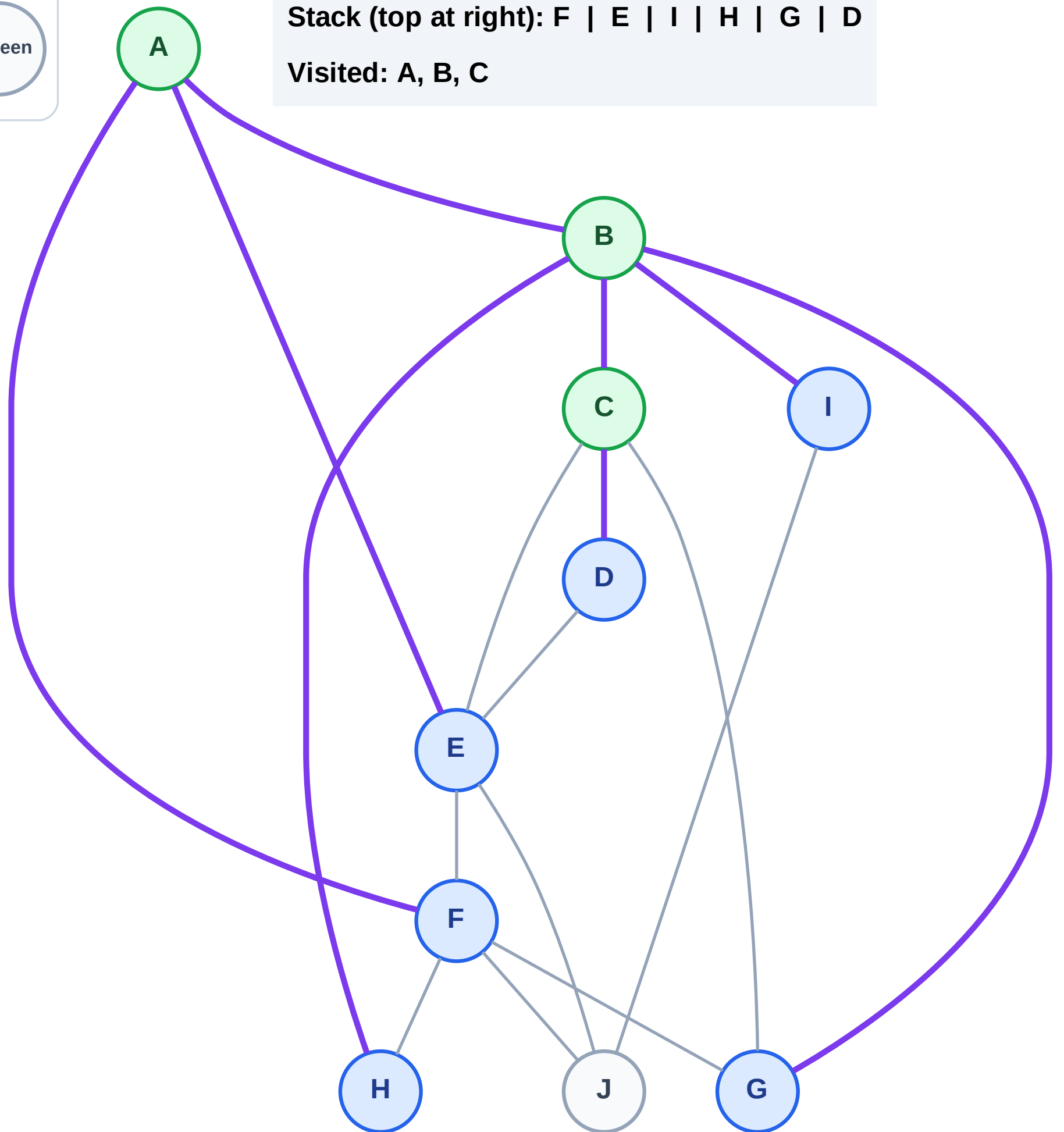
Step 7: Discover D from C

Legend



Stack (top at right): F | E | I | H | G | D

Visited: A, B, C



DFS Traversal

Step 8: Process node D

Legend

Complete

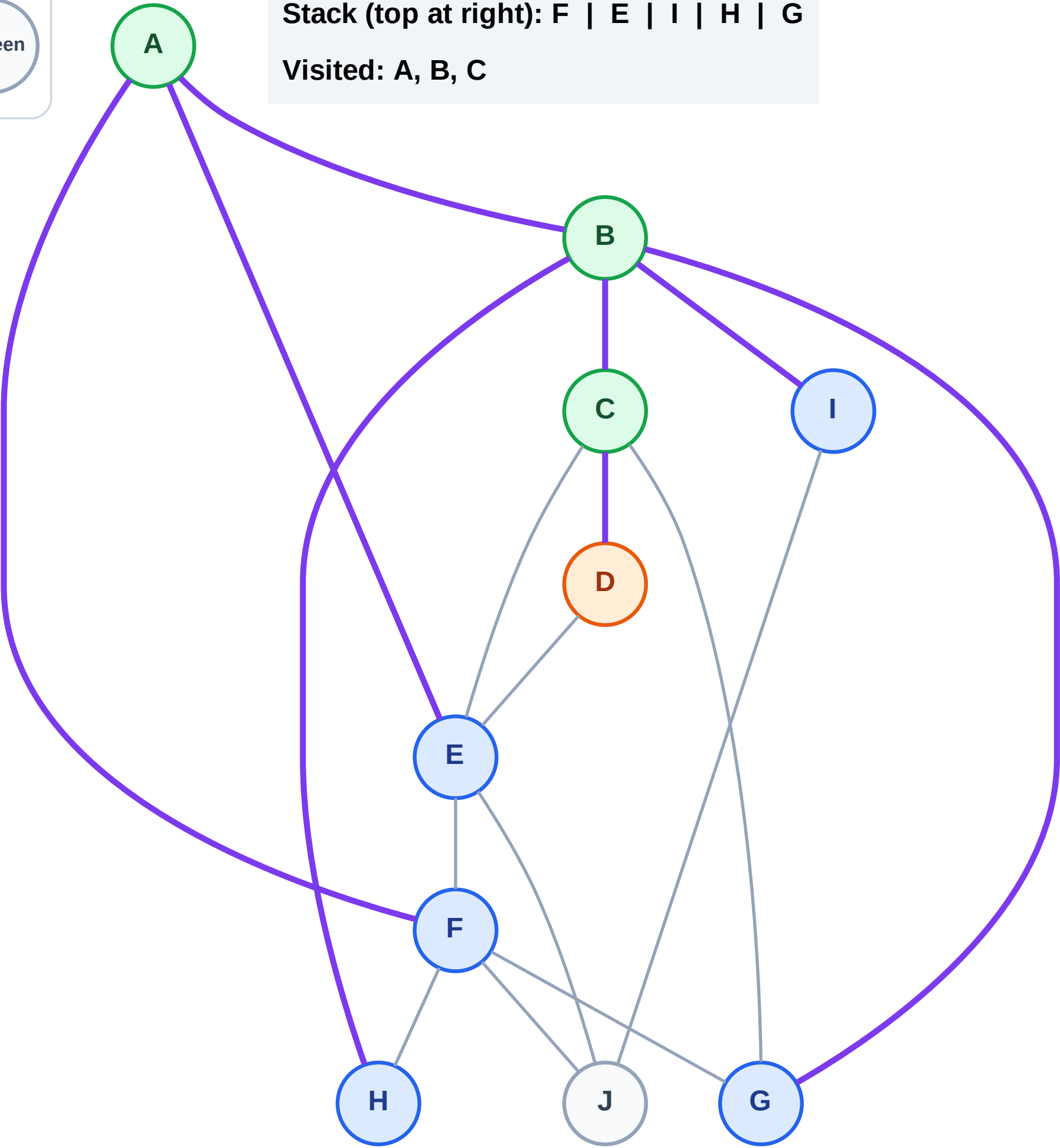
Processing

Discovered

Unseen

Stack (top at right): F | E | I | H | G

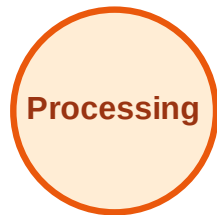
Visited: A, B, C



DFS Traversal

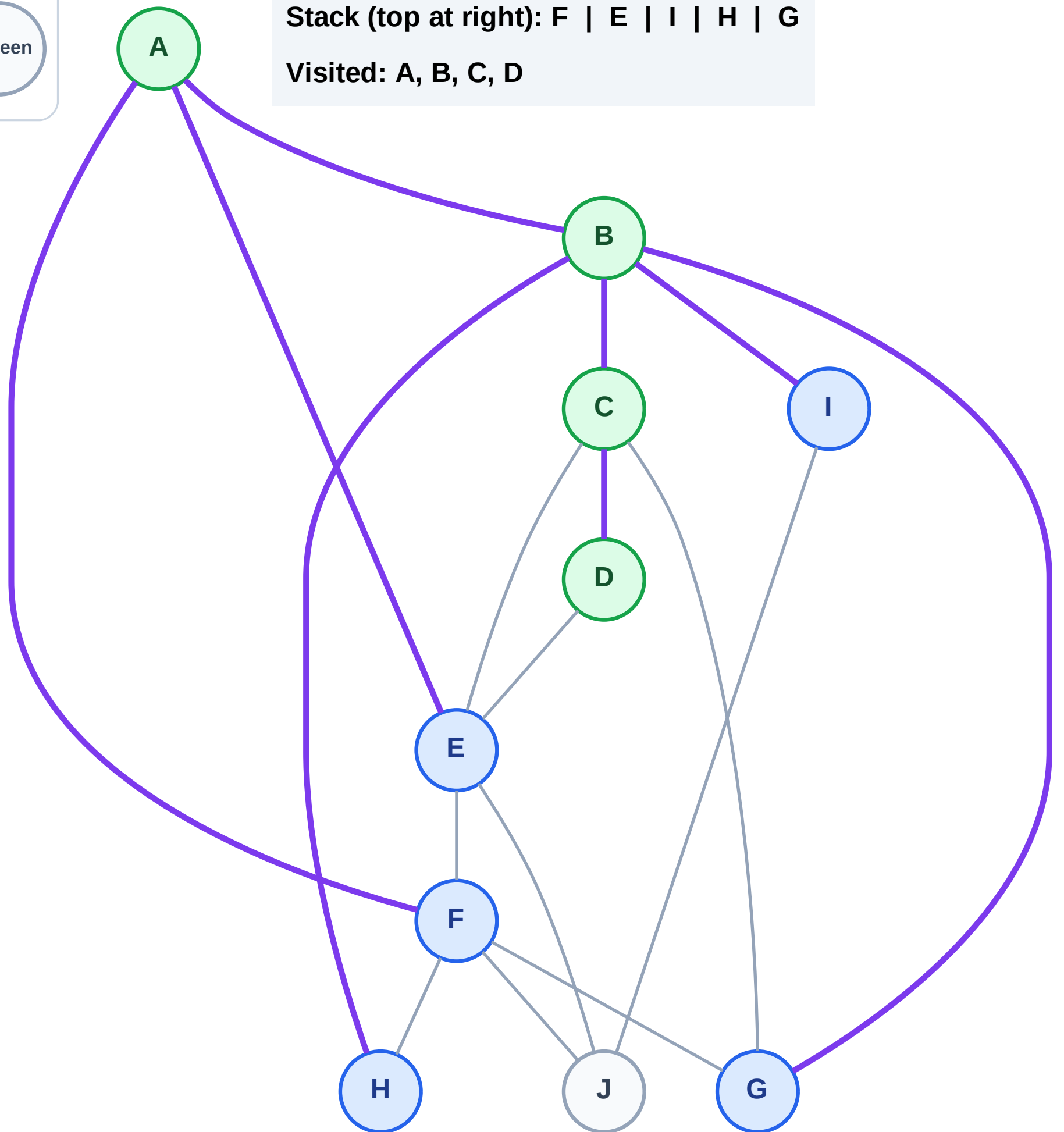
Step 9: No new neighbors from D

Legend



Stack (top at right): F | E | I | H | G

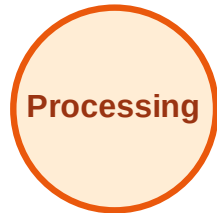
Visited: A, B, C, D



DFS Traversal

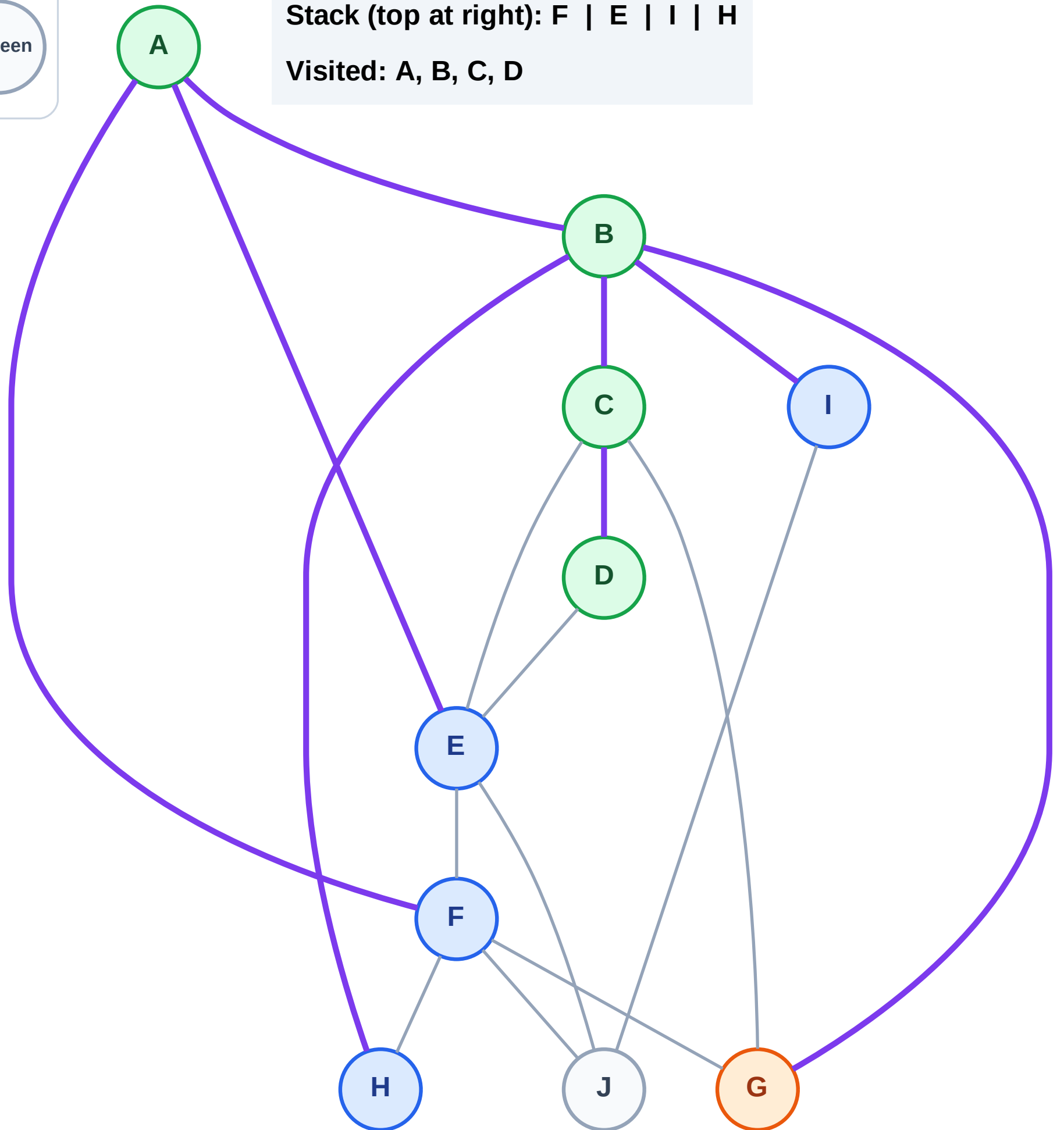
Step 10: Process node G

Legend



Stack (top at right): F | E | I | H

Visited: A, B, C, D



DFS Traversal

Step 11: No new neighbors from G

Legend

Complete

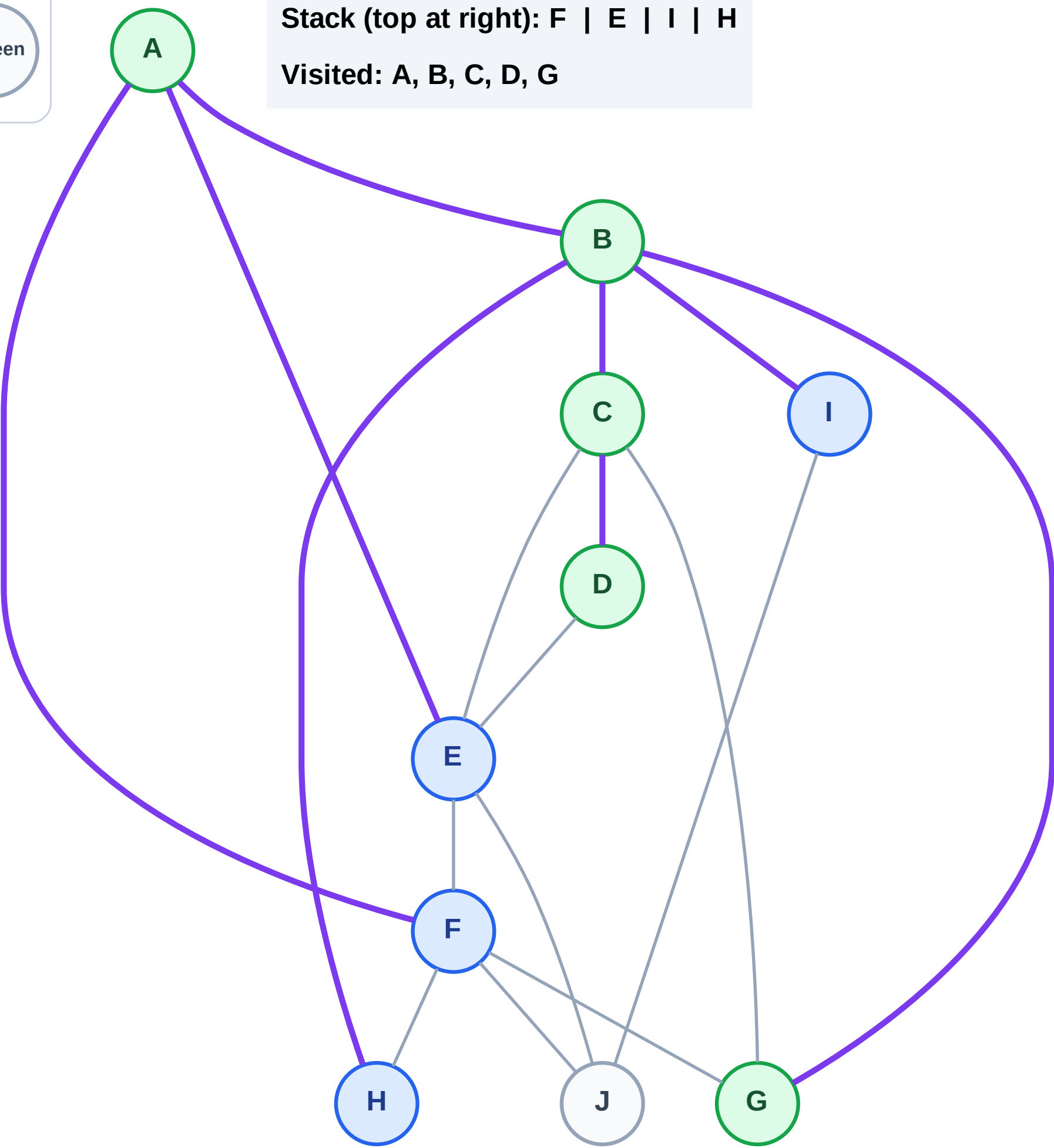
Processing

Discovered

Unseen

Stack (top at right): F | E | I | H

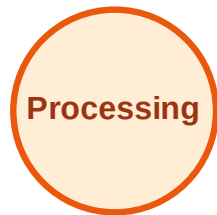
Visited: A, B, C, D, G



DFS Traversal

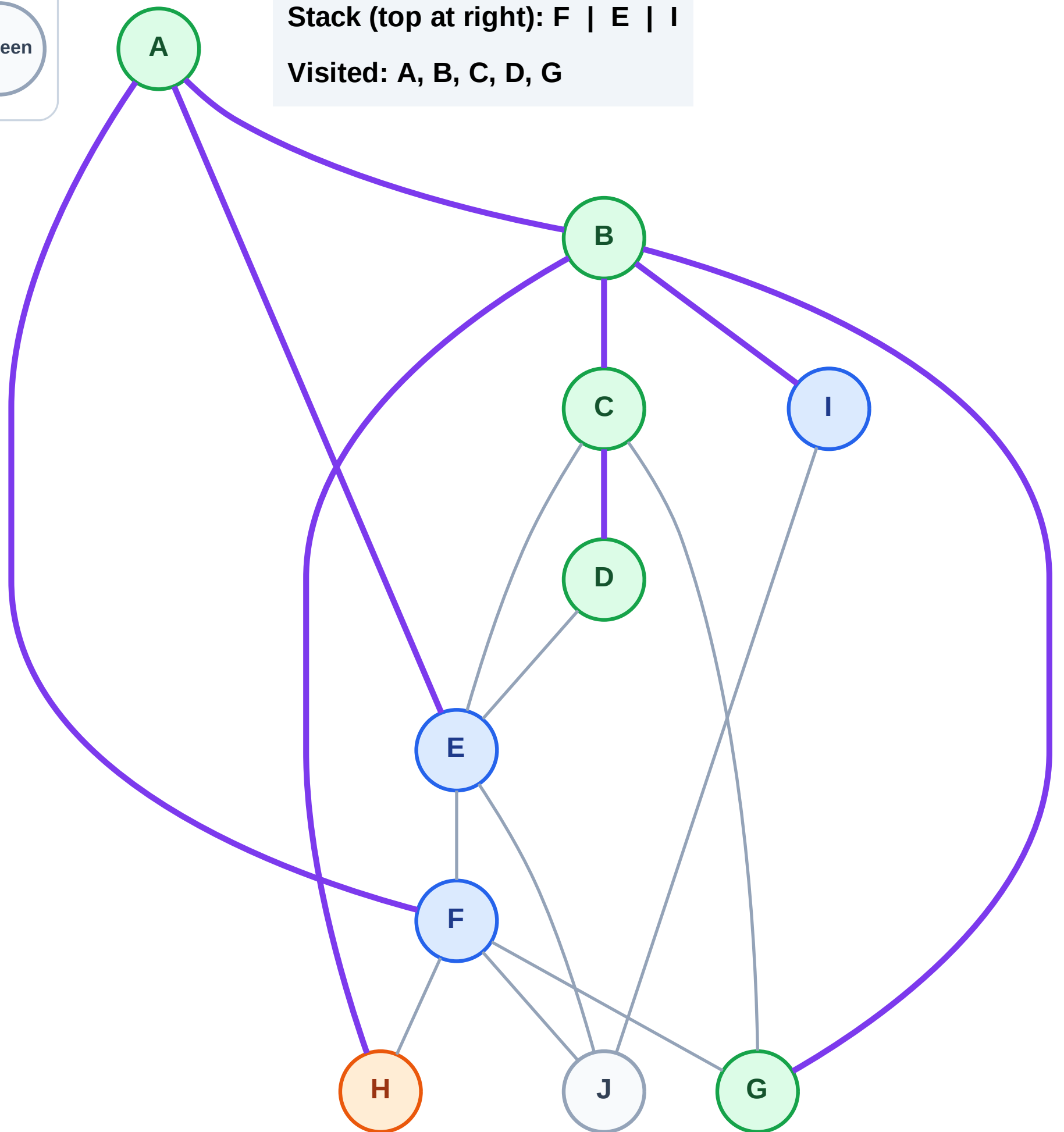
Step 12: Process node H

Legend



Stack (top at right): F | E | I

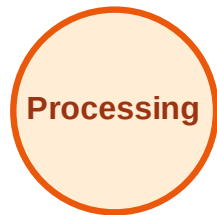
Visited: A, B, C, D, G



DFS Traversal

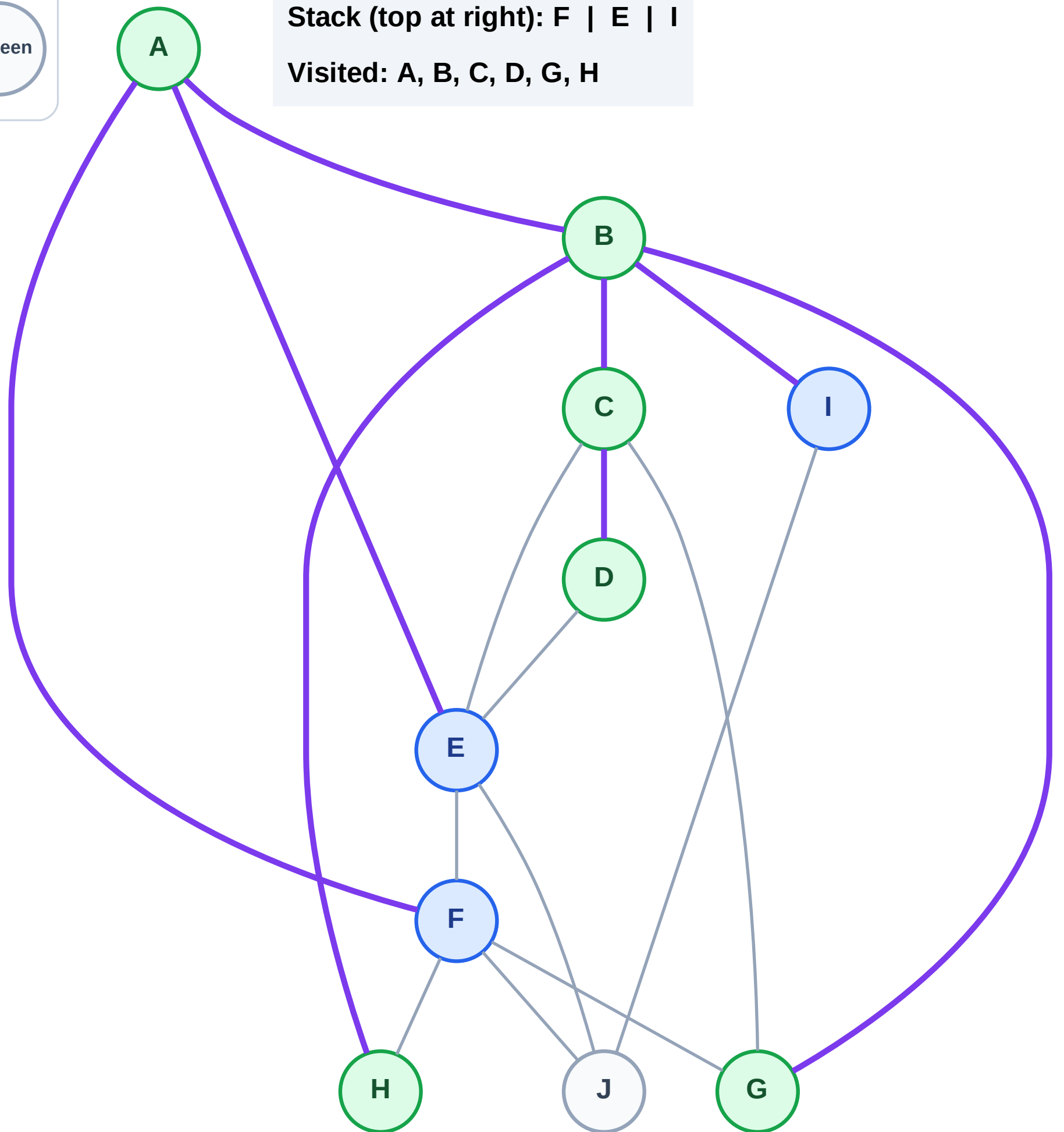
Step 13: No new neighbors from H

Legend



Stack (top at right): F | E | I

Visited: A, B, C, D, G, H



DFS Traversal

Step 14: Process node I

Legend

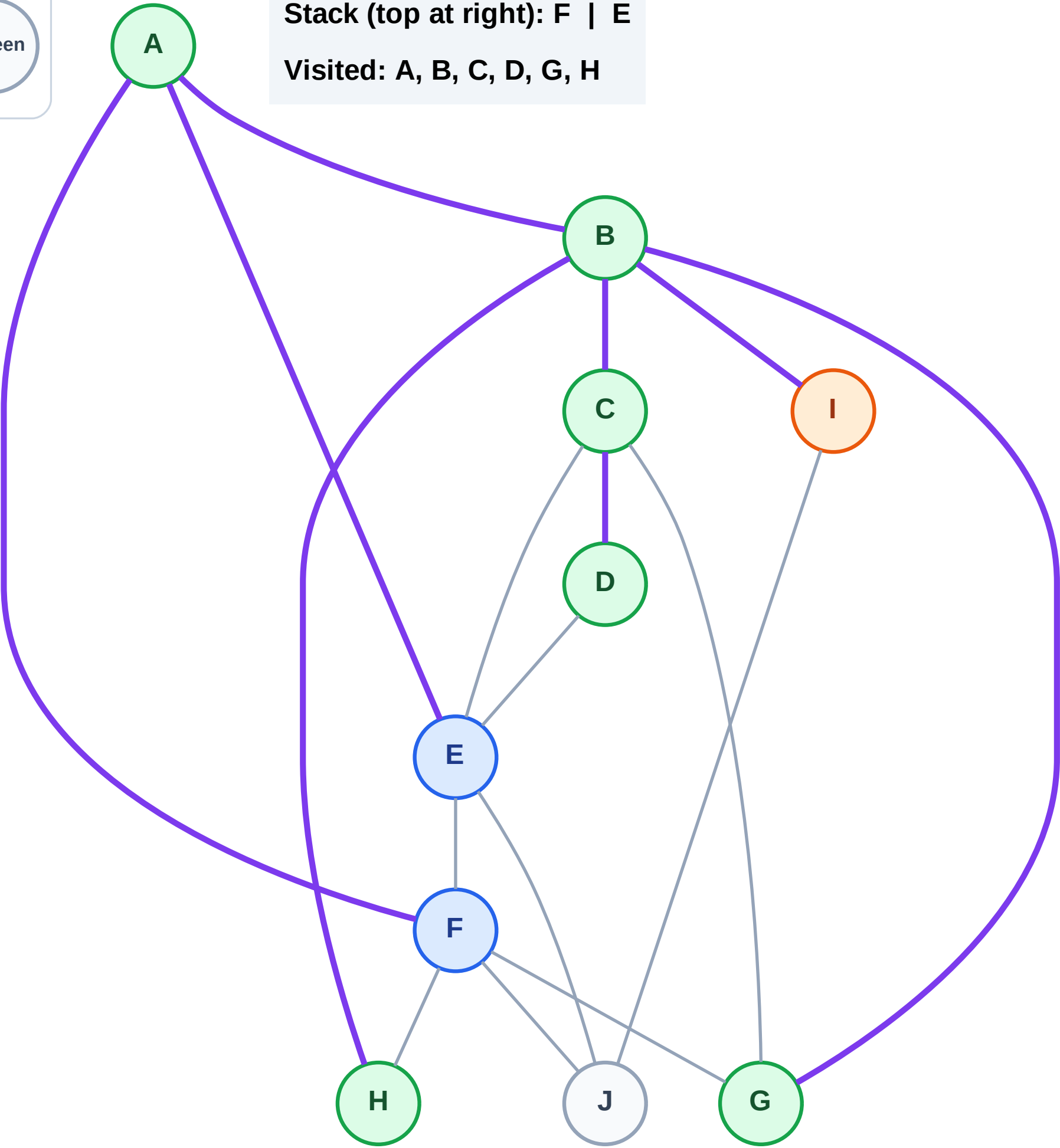
Complete

Processing

Discovered

Unseen

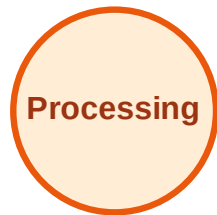
Stack (top at right): F | E
Visited: A, B, C, D, G, H



DFS Traversal

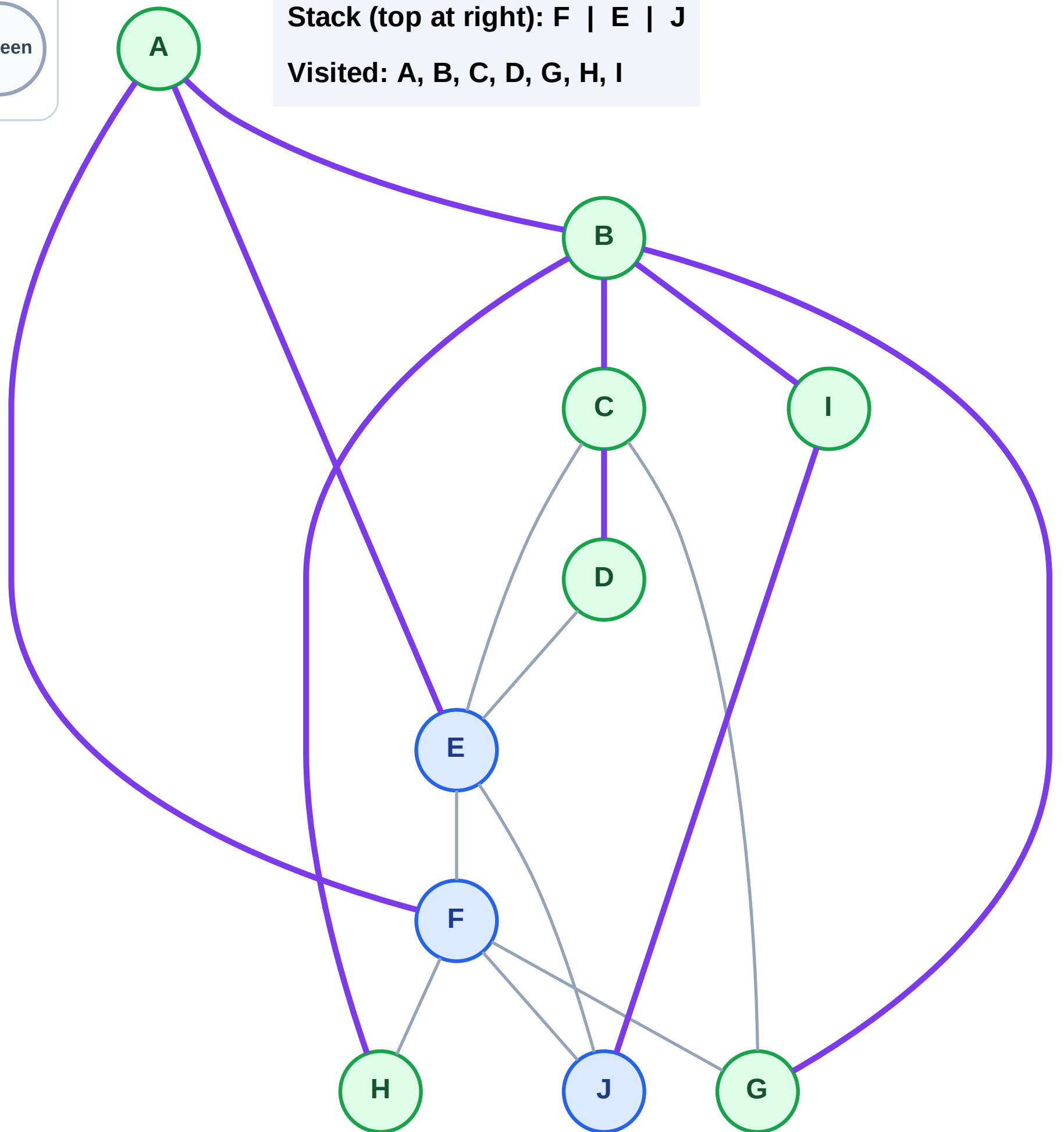
Step 15: Discover J from I

Legend

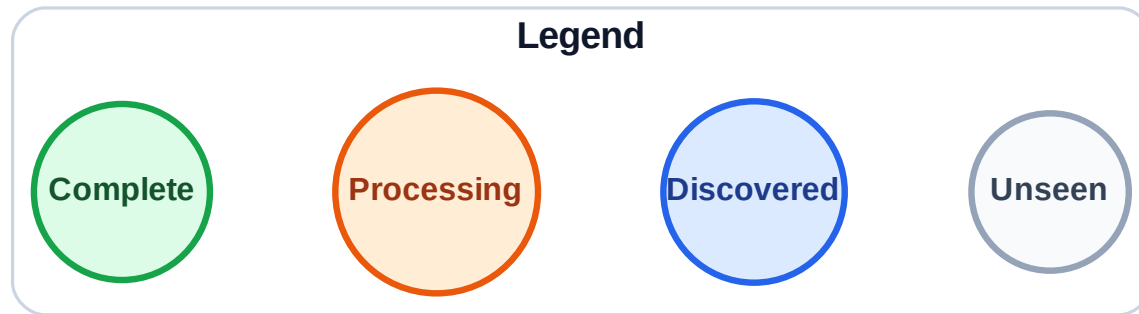


Stack (top at right): F | E | J

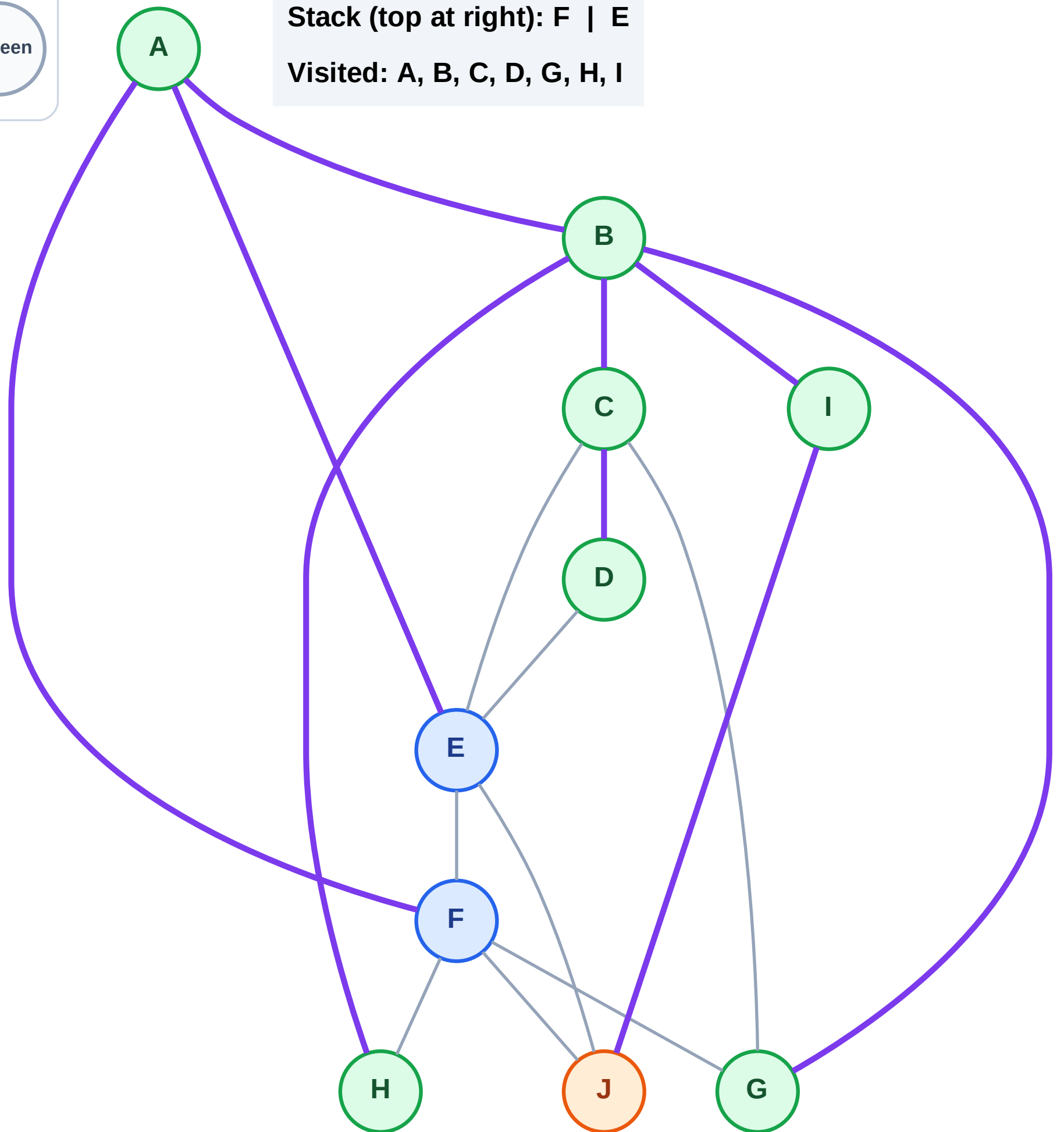
Visited: A, B, C, D, G, H, I



Step 16: Process node J



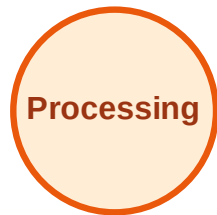
Visited: A, B, C, D, G, H, I



DFS Traversal

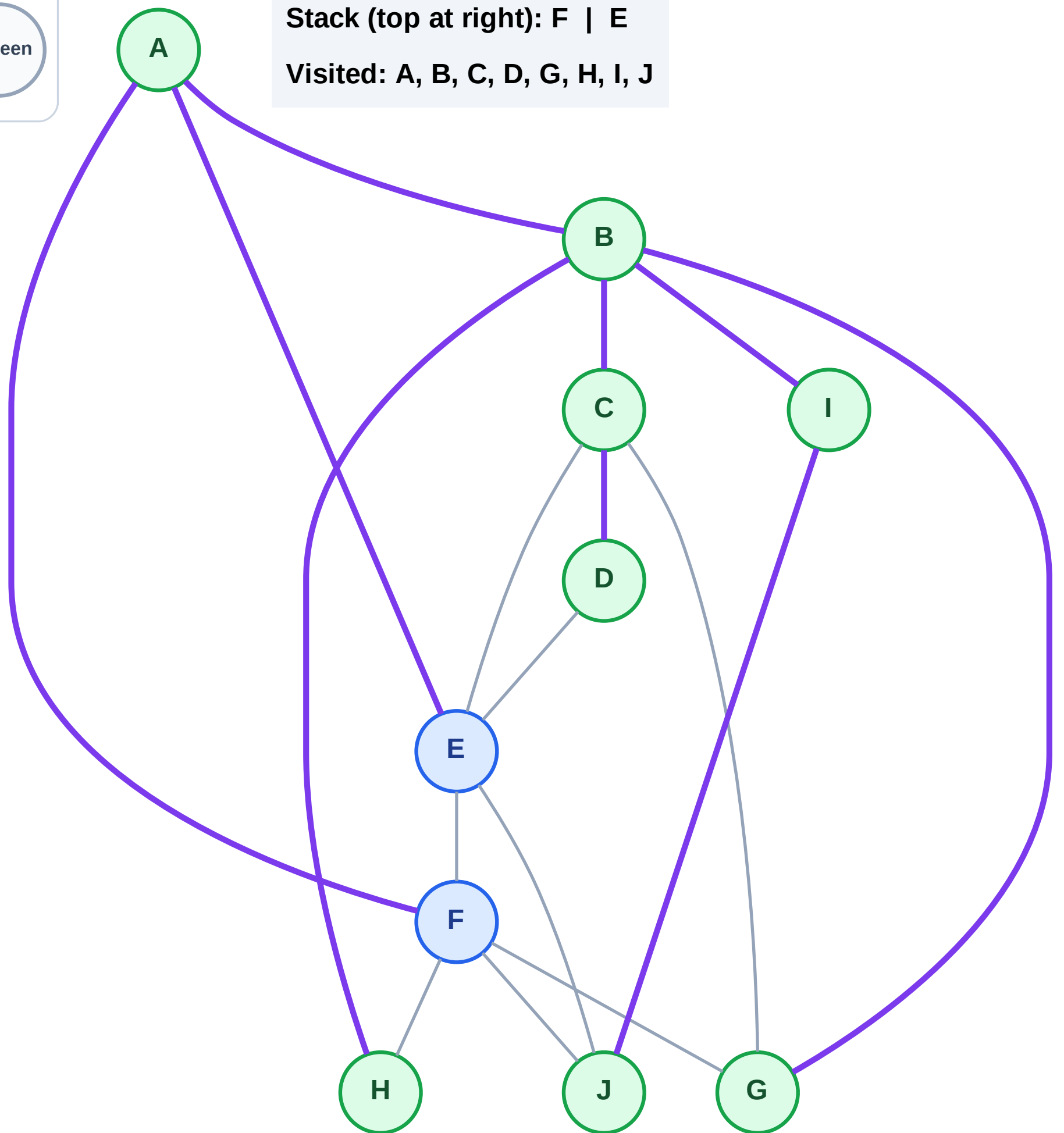
Step 17: No new neighbors from J

Legend



Stack (top at right): F | E

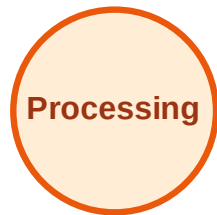
Visited: A, B, C, D, G, H, I, J



DFS Traversal

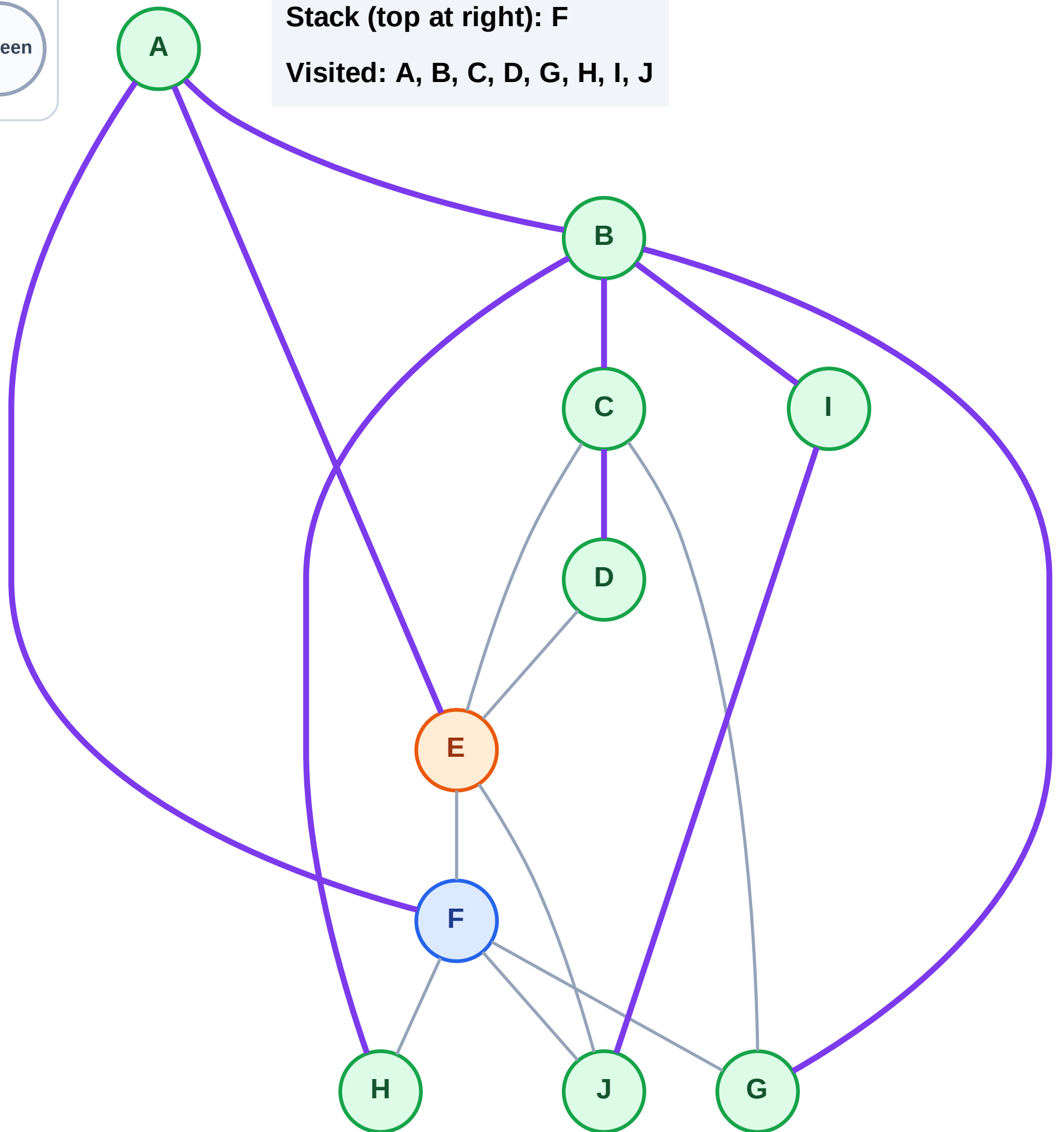
Step 18: Process node E

Legend



Stack (top at right): F

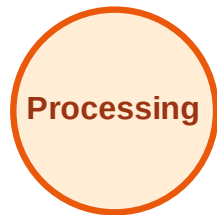
Visited: A, B, C, D, G, H, I, J



DFS Traversal

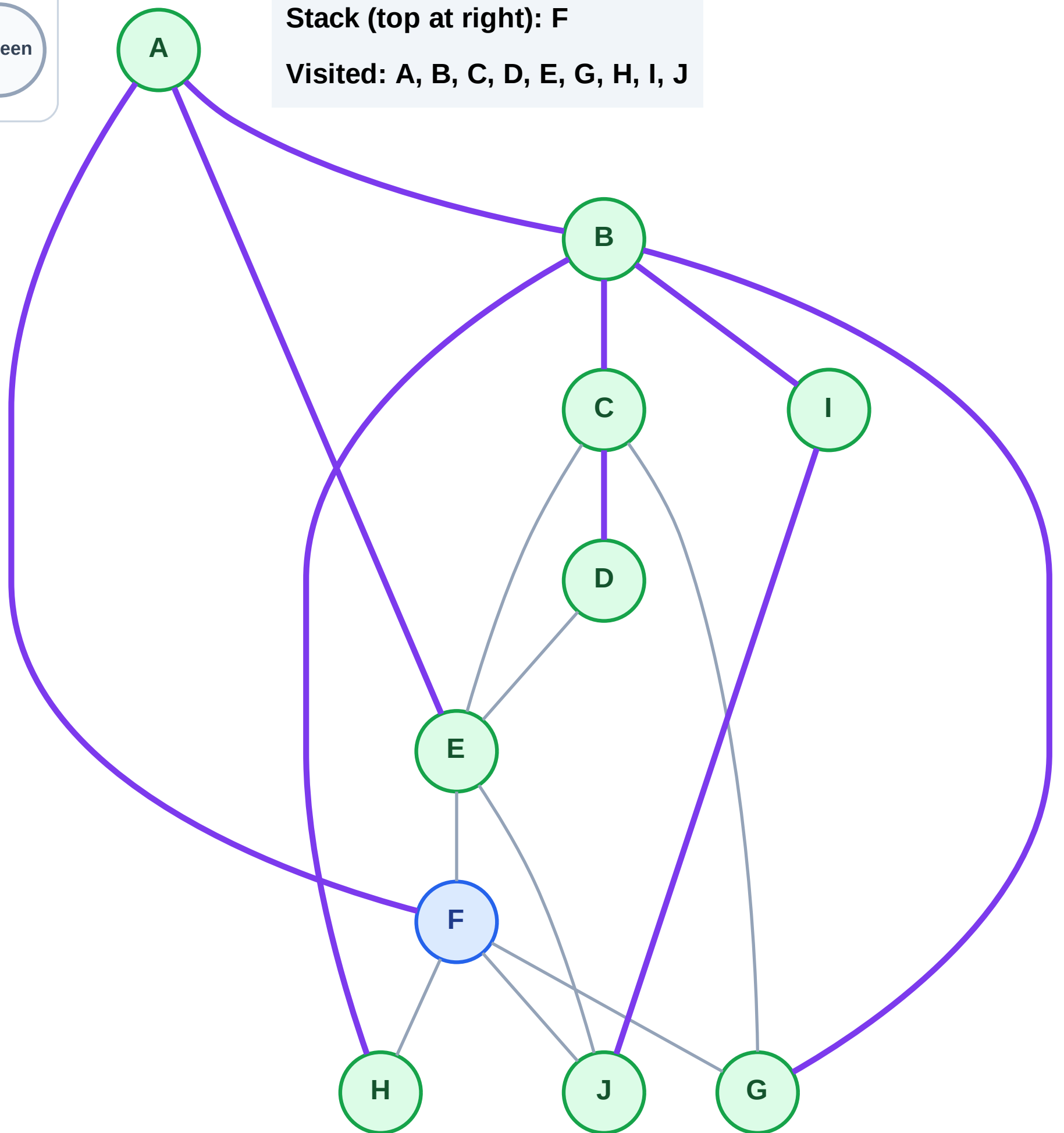
Step 19: No new neighbors from E

Legend



Stack (top at right): F

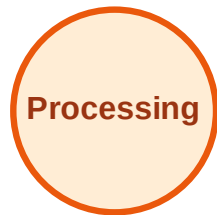
Visited: A, B, C, D, E, G, H, I, J



DFS Traversal

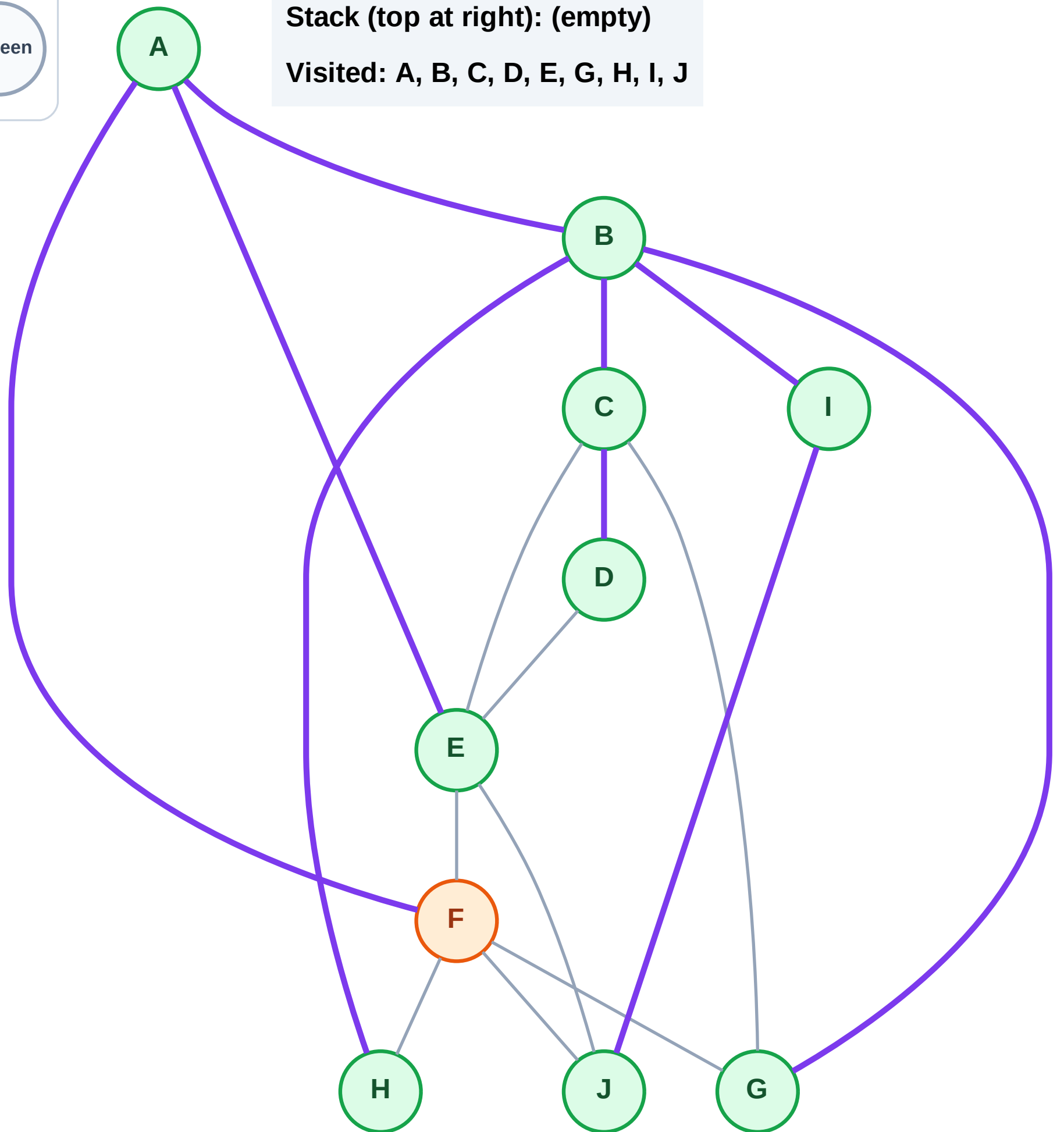
Step 20: Process node F

Legend



Stack (top at right): (empty)

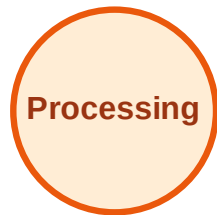
Visited: A, B, C, D, E, G, H, I, J



DFS Traversal

Step 21: No new neighbors from F

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

